

CHIRON v0.52alpha Manual and User Guide

Johan Bijnens

Department of Astronomy and Theoretical Physics, Lund University
Sölvegatan 14A, SE 22362 Lund, Sweden

Manual version of February 2015

Abstract

This manual and user guide describes the classes and functions contained in the ChPT program collection CHIRONv0.52alpha which includes the numerical library `jnumlib` and the ChPT routine library `chiron`.

This text is licensed under the creative commons license
CC-BY 4.0 (<http://creativecommons.org/licenses/by/4.0/>), except some parts which
are under their own license.

Contents

1	Introduction	3
2	Guidelines	4
2.1	Main comments	4
2.2	Some caution for use	4
3	Files, installation and testroutines	5
3.1	Files	5
3.2	Installation	5
3.3	testroutines	6
4	jbnunlib	6
4.1	Complex numbers	6
4.2	Special functions	6
4.2.1	jbdli2	6
4.2.2	Bessel functions	6
4.2.2.1	jbdbesi0	6
4.2.2.2	jbdbesi1	6
4.2.2.3	jbdbesk0	7
4.2.2.4	jbdbesk1	7
4.2.2.5	jbdbesk2	7
4.2.2.6	jbdbesk3	7
4.2.2.7	jbdbesk4	7
4.2.3	Theta and related functions	7
4.2.3.1	jbdtheta30	7
4.2.3.2	jbdtheta30m1	8
4.2.3.3	jbdtheta32	8
4.2.3.4	jbdtheta34	8
4.2.3.5	jbdtheta2d0	9
4.2.3.6	jbdtheta2d0m1	9
4.2.3.7	jbdtheta2d02	9
4.3	Integration routines	9
4.3.1	One dimension, real	9
4.3.1.1	jbdgauss	10
4.3.1.2	jbdgauss2	10
4.3.1.3	jbdquad15	10
4.3.1.4	jbdquad21	11
4.3.2	One dimension, real with singularity	11
4.3.2.1	jbdcauch	11
4.3.2.2	jbdcauch2	12

4.3.2.3	jbdsing15	12
4.3.2.4	jbdsing21	12
4.3.3	One dimension, complex	13
4.3.3.1	jbwgauss	13
4.3.3.2	jbwgauss2	13
4.3.3.3	jbwquad15	14
4.3.3.4	jbwquad21	14
4.3.4	Two dimensions, real	14
4.3.4.1	jbdad2	14
4.3.5	Three dimensions, real	15
4.3.5.1	jbdad3	15
5	Chiral Perturbation Theory	15
6	Data structures	15
6.1	Three flavour ChPT	16
6.1.1	Class: <code>physmass</code>	16
6.1.2	Class: <code>lomass</code>	16
6.1.3	Class: <code>quarkmass</code>	17
6.1.4	NLO LECs: Class <code>Li</code>	18
6.1.5	NLO LECs: Class <code>Ci</code>	19
7	Loop integrals	20
7.1	Tadpole or one-propagator integrals	21
7.2	Bubbles or two-propagator integrals	22
7.2.1	Definitions	22
7.2.2	Analytical implementation	23
7.2.3	Numerical implementation	23
7.3	Sunset integrals	24
7.3.1	Definition	24
7.3.2	Functions	25
7.4	Sunsetintegrals with different powers of propagators	26
7.4.1	Definition	26
7.4.2	Functions	27
7.5	Finite volume tadpole interals	27
7.5.1	Definitions	27
7.5.2	Functions	28
7.6	Finite volume sunsetintegrals	29
7.6.1	Definitions	29
7.6.2	Functions	30
7.7	Finite volume sunsetintegrals with different powers of propagators	32
7.7.1	Definitions	32
7.7.2	Functions	32

8	Three flavour isospin conserving results	34
8.1	Masses, decay constants and vacuum-expectation-values: in physical	34
8.1.1	Masses	34
8.1.2	Decay constants	35
8.1.3	<code>getfpimeta</code>	36
8.1.4	Vacuum-expectation-values	37
8.2	Masses, decay constants and vacuum-expectation-values: in lowest order .	38
8.2.1	Masses: in lowest order	38
8.2.2	Decay constants: in lowest order	40
8.2.3	Vacuum-expectation-values: in lowest order	41
8.3	Masses and decay constants at finite volume	42
8.3.1	Masses at finite volume	42
8.3.2	Decay constants at finite volume	44
A	GNU GENERAL PUBLIC LICENSE	45
B	Creative Commons Attribution 4.0 International Public License	52
	References	56
	Index	58

1 Introduction

This is the manual and user guide for the Chiral Perturbation Theory package `CHIRON` v0.52alpha. It also defines the functions included in a more extended fashion as compared to the published short description [1]. There is obviously a large overlap with that publication. The numerical routines are described in Sect. 4. The remaining sections are devoted to the `chiron` library.

This manual is released under the creative commons license CC-BY 4.0 [2] as reproduced in App. B except for the parts in App. A and App. B which have their own licenses. The software itself is released under the GNU General Public License (GPL) version 2 or later, which is reproduced in App. A.

Kheiron, $X\epsilon\iota\rho\omega\nu$, or Chiron, was the wisest and eldest of the Centaurs, half-horse men of Greek mythology. His name comes from the Greek word for hand (Kheir) which is also the origin of the word chiral which is his name was deemed appropriate for this package [3].

2 Guidelines

2.1 Main comments

Most of these routines were produced during and after scientific research. They are licensed under the GPL v2 or later, see App. A, [4] or the file `COPYING` in the main directory, so you have very strong rights in using and modifying them. However, please respect the guidelines as described in the file `GUIDELINES` in the main directory. A summary of these is

- Citations are important in the academic world so when using these please both cite the relevant `CHIRON` publication [1] and the papers where the work itself was done as quoted in the different chapters.
- Suspected bugs, proposed fixes and suggestions should preferably be communicated to the author(s) so they can be added in future releases.
- If you distribute modified versions, please indicate clearly the modifications in the source and at the point of distributions. However, the preferred way to introduce changes is via future releases.
- To make published results reproducible, the exact versions of the code that were used should be kept. This includes the values of all parameters used including the precisions.

2.2 Some caution for use

These routines have been used and tested in a ChPT environment using units in powers of GeV. Typical accuracies are set by default to relevant and obtainable values for that case. In addition, there are often special cases where the routines might not work, often due to 0/0 or large cancelations.

Similar comments apply to the special functions included. They are sufficiently accurate for the purposes they were used for originally and usually return values with a precision close to double precision but this is not guaranteed.

In some cases, the large formulas have inherently large cancelations. This might lead to degrading of precision in unexpected places. Use common (scientific) sense to judge the quality of the results.

Finally, there are a number of internal functions and extensions already present in the source code but not yet documented in this manual. These might change and have not been tested as well as the documented ones. In particular interfaces etc. might change.

3 Files, installation and testroutines

The package can be downloaded from [5]. There are ready to install libraries there for some cases, but in general it is better to compile it for your own system. C++ can have a large overhead in calling classes and functions compared to FORTRAN. Therefore always compile the library with optimization. The interfaces are as much as possible defined with the keyword `const` to allow the compiler to optimize more efficiently.

3.1 Files

The gzipped tarred file (`chiron.vvvv.tar.gz`) will produce a directory `chiron.vvvv` with a number of subdirectories. `vvvv` is version information. The created directory is called the main directory in the remainder.

The main directory contains the files `COPYING`, `INSTRUCTIONS`, `GUIDELINES` and a `Makefile`. The subdirectory contains the documentation. The latest published article about CHIRON, this manual (`manual.tex`), a list of files (`filelist.txt`) and a summary of things added since earlier versions (`Changelog.txt`).

The subdirectory `lib` will after compiling contain the compiled libraries `libjbnumlib.a` and `libchiron.a`.

The subdirectory `include` contains all the needed header files. The subdirectory `src` contains the source files. `test` contains the testing and example programs. `testoutputs` contains the output the testprograms should produce.

Typically for each subject `xxx` there are files `xxx.h`, `xxx.cc`, `testxxx.cc` and `testxxx.dat` in the respective directories.

There are a few extra files around as well. These typically contain inputs needed or large sets of constants.

3.2 Installation

The main steps are to run `make` in the main directory. This should produce the files `libjbnumlib.a` and `libchiron.a` and also copy them to the `lib` subdirectory. You might have to change the variables `CC`, `CFLAGS` and `CFLAGTESTS`. `CC` should specify the C++ compiler and the options to be used for everything. `CFLAGS` can be used to specify additional options in compiling the libraries and `CFLAGTESTS` to specify additional options for the testing programs.

“make clean” can be used to remove many of the files created during compiling.

The actual installation is by putting the contents of the `include` directory somewhere in the include path of your compiler and the two files `libjbnumlib.a` and `libchiron.a` somewhere in the library path. For many C++ compilers the paths are given in the environment variables `CPLUS_INCLUDE_PATH` and `LIBRARY_PATH` respectively.

3.3 testroutines

For every file `xxx.h` and `xxx.cc` included for `chiron` there is a testing/example code `testxxx.cc` in the subdirectory `test`. These can be compiled using “make testxxx” in the main directory. Executing the resulting file `a.out` should then produce output identical (up to the precision specified and possible randomly generated cases) to the file `testxxx.dat` in the subdirectory `testoutputs`.

4 jbnunlib

4.1 Complex numbers

Complex numbers are defined via the standard C++ library and an abbreviation provided as

```
typedef std::complex<double> dcomplex;
```

All variables declared complex will be of the this type and referred to as `dcomplex` in the remainder.

4.2 Special functions

4.2.1 jbdli2

```
dcomplex jbdli2(const dcomplex x)
```

Returns the complex dilogarithm or Spence function defined by

$$\mathrm{Li}_2(x) = - \int_0^1 dt \frac{\log(1 - xt)}{t}, \quad (1)$$

where it converges and analytic continuation. Cut defined on the positive real axis from 1 to ∞ . Uses the properties of the dilogarithm to transform the argument and then the Bernouilly series as described in [6].

Defined in `jbnunlib.h` and implemented in `jbdli2.cc`.

4.2.2 Bessel functions

4.2.2.1 jbdbesi0

```
double jbdbesi0(const double x)
```

Returns the modified Bessel function I_0 for real values of the argument. A simple port to C++ of CERNLIB[7] routine DBESI0.

Defined in `jbnunlib.h`, implemented in `jbdbesik.cc`.

4.2.2.2 jbdbesi1

```
double jbdbesi1(const double x)
```

Returns the modified Bessel function I_1 for real values of the argument. A simple port to C++ of CERNLIB[7] routine DBESI1.

Defined in `jbnunlib.h`, implemented in `jbdbesik.cc`.

4.2.2.3 `jbdbesk0`

`double jbdbesk0(const double x)`

Returns the modified Bessel function K_0 for real values of the argument. A simple port to C++ of CERNLIB[7] routine DBESK0.

Defined in `jbnunlib.h`, implemented in `jbdbesik.cc`.

4.2.2.4 `jbdbesk1`

`double jbdbesk1(const double x)`

Returns the modified Bessel function K_1 for real values of the argument. A simple port to C++ of CERNLIB[7] routine DBESK1.

Defined in `jbnunlib.h`, implemented in `jbdbesik.cc`.

4.2.2.5 `jbdbesk2`

`double jbdbesk2(const double x)`

Returns the modified Bessel function K_2 for real values of the argument. Uses the recursion relations for Bessel functions and `jbdbesk0` and `jbdbesk1`.

Defined in `jbnunlib.h`, implemented in `jbdbesik.cc`.

4.2.2.6 `jbdbesk3`

`double jbdbesk3(const double x)`

Returns the modified Bessel function K_3 for real values of the argument. Uses the recursion relations for Bessel functions and `jbdbesk0` and `jbdbesk1`.

Defined in `jbnunlib.h`, implemented in `jbdbesik.cc`.

4.2.2.7 `jbdbesk4`

`double jbdbesk4(const double x)`

Returns the modified Bessel function K_4 for real values of the argument. Uses the recursion relations for Bessel functions and `jbdbesk0` and `jbdbesk1`.

Defined in `jbnunlib.h`, implemented in `jbdbesik.cc`.

4.2.3 Theta and related functions

4.2.3.1 `jbdtheta30`

`double jbdtheta30(const double q)`

Returns the value of the function

$$\theta_{30}(q) = 1 + 2 \sum_{n=1,\infty} q^{(n^2)} = \sum_{n=-\infty,\infty} q^{(n^2)}. \quad (2)$$

This function is related to the third Jacobi theta function. For small q the summation in (2) is used directly. For larger q the identity

$$\theta_{30}(q) = \sqrt{\frac{\lambda}{\pi}} \theta_{30}(e^{-\lambda}) \quad (3)$$

with $\lambda = \pi^2/|\log(q)|$ is used instead. This is related to the modular invariance for the higher dimensional case. Precision can be judged by comparing the two series to each other. Same idea as used in the CERNLIB[7] routine `DTHETA`.

Defined in `jbnlib.h`, implemented in `jbdtheta30.cc`.

4.2.3.2 jbdtheta30m1

`double jbdtheta30m1(const double q)`

Returns the value of the function

$$\theta_{30}(q) - 1 = 2 \sum_{n=1,\infty} q^{(n^2)} = \sum_{n \in \mathbb{Z}, n \neq 0} q^{(n^2)}. \quad (4)$$

Implementation as for `jbdtheta30` but without the 1. Especially for small q often needed to keep accuracy in the finite volume applications in ChPT.

Defined in `jbnlib.h`, implemented in `jbdtheta30.cc`.

4.2.3.3 jbdtheta32

`double jbdtheta32(const double q)`

Returns the value of the function

$$\theta_{32}(q) = 2 \sum_{n=1,\infty} q^{(n^2)} = \sum_{n=-\infty,\infty} n^2 q^{(n^2)} = q \frac{d}{dq} \theta_{30}(q). \quad (5)$$

For small q the summation in (5) is used directly. For larger q the derivative of the right-hand-side of the identity (3) is used.

Defined in `jbnlib.h`, implemented in `jbdtheta32.cc`.

4.2.3.4 jbdtheta34

`double jbdtheta34(const double q)`

Returns the value of the function

$$\theta_{34}(q) = 2 \sum_{n=1,\infty} n^4 q^{(n^2)} = \sum_{n=-\infty,\infty} n^4 q^{(n^2)} = \left(q \frac{d}{dq} \right)^2 \theta_{30}(q). \quad (6)$$

For small q the summation in (6) is used directly. For larger q the appropriate derivative of the right-hand-side of the identity (3) is used.

Defined in `jbnlib.h`, implemented in `jbdtheta34.cc`.

4.2.3.5 jbdtheta2d0

`double jbdtheta2d0(const double a, const double b, const double c)`

Returns the value of the function

$$\theta_0^{(2)}(a, b, c) = \sum_{n_1, n_2 = -\infty, \infty} e^{-an_1^2 - bn_2^2 - c(n_1 - n_2)^2}. \quad (7)$$

There are many higher-dimensional generalizations of the Jacobi theta functions. The modular invariance properties of these are discussed in App. B of [8] and are used in the evaluation to speed up the calculation. It should be noted that $\theta_0^{(2)}(a, b, c)$ is fully symmetric in a, b, c .

Defined in `jbnlib.h` and implemented in `jbdtheta2d0.cc`.

4.2.3.6 jbdtheta2d0m1

`double jbdtheta2d0(const double a, const double b, const double c)`

Returns the value of the function

$$\theta_0^{(2)}(a, b, c) - 1 = \sum_{n_1, n_2 = -\infty, \infty} e^{-an_1^2 - bn_2^2 - c(n_1 - n_2)^2} - 1 = \sum_{\substack{n_1, n_2 \in \mathbb{Z} \\ (n_1, n_2) \neq (0, 0)}} e^{-an_1^2 - bn_2^2 - c(n_1 - n_2)^2} - 1. \quad (8)$$

Method as in `jbdtheta2d0` but the 1 removed, more accurate for small a, b, c as often needed in finite volume ChPT.

Defined in `jbnlib.h` and implemented in `jbdtheta2d0m.cc`.

4.2.3.7 jbdtheta2d02

`double jbdtheta2d02(const double a, const double b, const double c)`

Returns the value of the function

$$\theta_{02}^{(2)}(a, b, c) = \sum_{n_1, n_2 = -\infty, \infty} n_1^2 e^{-an_1^2 - bn_2^2 - c(n_1 - n_2)^2} = -\frac{\partial}{\partial a} \theta_0^{(2)}(a, b, c). \quad (9)$$

It should be noted that $\theta_{02}^{(2)}(a, b, c)$ is symmetric in b, c . Method similar to `jbdtheta2d0`. Defined in `jbnlib.h` and implemented in `jbdtheta2d02.cc`.

4.3 Integration routines

4.3.1 One dimension, real

The interface of these routines is identical so they can be simply interchanged. For most problems the speed decreases as `jbdquad15`, `jbdquad21`, `jbdgauss2`, `jbdgauss` but this is

somewhat dependent on the function integrated and the precision requested.

The routines do not use the endpoints so an integrable singularity at the endpoint can be done but an integrand transformation that removes the singularity will lead to a much better performance.

An example program that shows the relative speeds is in `testintegralsreal.cc`.

4.3.1.1 jbdgauss

`double jbdgauss(f,a,b,eps)`

`f: double (*f)(const double x)` The double precision function to be integrated over.

`a,b,eps: const double`

`a`: Lower limit of integration.

`b`: Upper limit of integration.

`eps`: Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Subroutine translated from the CERNLIB[7] routine `DGAUSS`. Uses 8 and 16 point Gaussian rules with the 16 point for the estimate and the difference for the error estimate.

Adaptive with a subdivision strategy.

Defined in `jbnlib.h`, implemented in `jbdfauss.cc`.

4.3.1.2 jbdgauss2

`double jbdgauss2(f,a,b,eps)`

`f: double (*f)(const double x)` The double precision function to be integrated over.

`a,b,eps: const double`

`a`: Lower limit of integration.

`b`: Upper limit of integration.

`eps`: Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Uses 8 and 16 point Gaussian rules with the 16 point for the estimate and the difference for the error estimate. Adaptive with a subdivision strategy. Very similar to `jbdfauss` but the subdivision strategy is more appropriate for high precision.

Defined in `jbnlib.h`, implemented in `jbdfauss2.cc`.

4.3.1.3 jbdquad15

`double jbdquad15(f,a,b,eps)`

`f: double (*f)(const double x)` The double precision function to be integrated over.

`a,b,eps: const double`

`a`: Lower limit of integration.

`b`: Upper limit of integration.

`eps`: Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Uses 15 point Gauss-Kronrod rule for the estimate and the difference with the embedded 7 point Gauss rule for the error estimate. Adaptive with a subdivision strategy appropriate for high precision.

Defined in `jbnlib.h`, implemented in `jbdquad15.cc`.

4.3.1.4 jbdquad21

`double jbdquad21(f,a,b,eps)`

`f: double (*f)(const double x)` The double precision function to be integrated over.

`a,b,eps: const double`

`a`: Lower limit of integration.

`b`: Upper limit of integration.

`eps`: Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Uses 21 point Gauss-Kronrod rule for the estimate and the difference with the embedded 10 point Gauss rule for the error estimate. Adaptive with a subdivision strategy appropriate for high precision.

Defined in `jbnlib.h`, implemented in `jbdquad21.cc`.

4.3.2 One dimension, real with singularity

The interface of these routines is identical so they can be simply interchanged. For most problems the speed decreases as `jbdquad15` or `jbdquad21`, `jbdgauss2`, `jbdgauss` but this is somewhat dependent on the function integrated and the precision requested.

An example program that shows the relative speeds is in `testintegralsrealsingular.cc`.

4.3.2.1 jbdcauch

`double jbdcauch(f,a,b,s,eps)`

`f: double (*f)(const double x)` The double precision function to be integrated over.

`a,b,s,eps: const double`

`a`: Lower limit of integration.

`b`: Upper limit of integration.

`s`: Place of the singularity.

`eps`: Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Subroutine translated from the CERNLIB[7] routine `DCAUCH`. Uses 8 and 16 point Gaussian rules with the 16 point for the estimate and the difference for the error estimate. Adaptive with a subdivision strategy. Integrates symmetrically around the singularity so it returns the integral in the sense of the principal value prescription. Uses `jbdgauss`.

Defined in `jbnlib.h`, implemented in `jbdcauch.cc`.

4.3.2.2 jbdcauch2

`double jbdcauch2(f,a,b,s,eps)`

`f: double (*f)(const double x)` The double precision function to be integrated over.

`a,b,s,eps: const double`

`a`: Lower limit of integration.

`b`: Upper limit of integration.

`s`: Place of the singularity.

`eps`: Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Subroutine translated from the CERNLIB[7] routine DCAUCH. Uses 8 and 16 point Gaussian rules with the 16 point for the estimate and the difference for the error estimate. Adaptive with a subdivision strategy more suitable for high precision. Integrates symmetrically around the singularity so it returns the integral in the sense of the principal value prescription. Uses `jbdgauss2`.

Defined in `jbnlib.h`, implemented in `jbdcauch2.cc`.

4.3.2.3 jbdsing15

`double jbdsing15(f,a,b,s,eps)`

`f: double (*f)(const double x)` The double precision function to be integrated over.

`a,b,s,eps: const double`

`a`: Lower limit of integration.

`b`: Upper limit of integration.

`s`: Place of the singularity.

`eps`: Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Subroutine similar to `jbdcauch2` but uses a Gauss-Kronrod 15 point rule for the estimate and the difference with the embedded 7 point Gauss rule for the error estimate. Adaptive with a subdivision strategy more suitable for high precision. Integrates symmetrically around the singularity so it returns the integral in the sense of the principal value prescription. Uses `jbdquad15`.

Defined in `jbnlib.h`, implemented in `jbdsing15.cc`.

4.3.2.4 jbdsing21

`double jbdsing21(f,a,b,s,eps)`

`f: double (*f)(const double x)` The double precision function to be integrated over.

`a,b,s,eps: const double`

`a`: Lower limit of integration.

`b`: Upper limit of integration.

`s`: Place of the singularity.

`eps`: Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Subroutine similar to `jbdcauch2` but uses a Gauss-Kronrod 21 point rule for the estimate and the difference with the embedded 10 point Gauss rule for the error estimate. Adaptive with a subdivision strategy more suitable for high precision. Integrates symmetrically around the singularity so it returns the integral in the sense of the principal value prescription. Uses `jbdquad21`.

Defined in `jbnlib.h`, implemented in `jbdsing21.cc`.

4.3.3 One dimension, complex

The interface of these routines is identical so they can be simply interchanged. For most problems the speed decreases as `jbwquad15` or `jbwquad21` or `jbwgauss2`, `jbwgauss` but this is somewhat dependent on the function integrated and the precision requested.

An example program that shows the relative speeds is in `testintegralscomplex.cc`.

4.3.3.1 jbwgauss

`dcomplex jbwgauss(f,a,b,eps)`

`f: dcomplex (*f)(const dcomplex x)` The complex double precision function to be integrated over.

`a,b: const dcomplex`

`a`: Lower endpoint of integration.

`b`: Upper endpoint of integration.

`eps: const double` Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Subroutine translated from the CERNLIB[7] routine `WGAUSS`. Uses 8 and 16 point Gaussian rules with the 16 point for the estimate and the difference for the error estimate. Adaptive with a subdivision strategy. The integration is the lineintegral over the straight line between `a` and `b`.

Defined in `jbnlib.h`, implemented in `jbwgauss.cc`.

4.3.3.2 jbwgauss2

`dcomplex jbwgauss2(f,a,b,eps)`

`f: dcomplex (*f)(const dcomplex x)` The complex double precision function to be integrated over.

`a,b: const dcomplex`

`a`: Lower endpoint of integration.

`b`: Upper endpoint of integration.

`eps: const double` Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Subroutine translated from the CERNLIB[7] routine `WGAUSS`. Uses 8 and 16 point Gaussian rules with the 16 point for the estimate and the difference for the error estimate. Adaptive with a subdivision strategy better suited for high precision. The integration is the lineintegral over the straight line between `a` and `b`.

Defined in `jbnumlib.h`, implemented in `jbwgauss2.cc`.

4.3.3.3 jbwquad15

`dcomplex jbwquad15(f,a,b,eps)`

`f: dcomplex (*f)(const dcomplex x)` The complex double precision function to be integrated over.

`a,b: const dcomplex`

`a`: Lower endpoint of integration.

`b`: Upper endpoint of integration.

`eps: const double` Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Subroutine similar to `jbwgauss2` but uses a 15 point Gauss-Kronrod rule for the estimate and the difference with the embedded 7 point Gauss rule for the error estimate. Adaptive with a subdivision strategy better suited for high precision. The integration is the lineintegral over the straight line between `a` and `b`.

Defined in `jbnumlib.h`, implemented in `jbwquad15.cc`.

4.3.3.4 jbwquad21

`dcomplex jbwquad21(f,a,b,eps)`

`f: dcomplex (*f)(const dcomplex x)` The complex double precision function to be integrated over.

`a,b: const dcomplex`

`a`: Lower endpoint of integration.

`b`: Upper endpoint of integration.

`eps: const double` Precision attempted to be reached: relative precision if absolute value of the integral is above 1, otherwise absolute precision.

Subroutine similar to `jbwgauss2` but uses a 21 point Gauss-Kronrod rule for the estimate and the difference with the embedded 10 point Gauss rule for the error estimate. Adaptive with a subdivision strategy better suited for high precision. The integration is the lineintegral over the straight line between `a` and `b`.

Defined in `jbnumlib.h`, implemented in `jbwquad21.cc`.

4.3.4 Two dimensions, real

4.3.4.1 jbdad2

`double jbdad2(f,a,b,relaps, relerr, ifail)`

`f: double (*f)(double x[])` The double precision function to be integrated over, `x[0]` and `x[1]` contain the values of the two variables to be integrated over.

`a: double a[]` `a[0]` and `a[1]` are the lower limits of integration.

`b: double b[]` `b[0]` and `b[1]` are the upper limits of integration.

`relaps: const double` requested relative precision of the integral.

`relerr`: double & returns the obtained relative precision via a reference.

`ifail`: int & returns an integer. Zero indicates success, if not zero the routine did not obtain the requested precision..

The function does a two dimensional integration over a hypercube. The underlying routine is `jbdadm1` which is a simple port to C++ of the CERNLIB[7] routine `DADMUL`. This in turn was based on [9].

Defined in `jbnulib.h`, implemented in `jbdadm1.cc`.

4.3.5 Three dimensions, real

4.3.5.1 `jbdad3`

`double jbdad3(f,a,b,relaps, relerr, ifail)`

`f`: double (*f)(double x[]) The double precision function to be integrated over, `x[0]`, `x[1]` and `x[2]` contain the values of the three variables to be integrated over.

`a`: double `a[]` `a[0]`, `a[1]` and `a[2]` are the lower limits of integration.

`b`: double `b[]` `b[0]`, `b[1]` and `b[2]` are the upper limits of integration.

`relaps`: const double requested relative precision of the integral.

`relerr`: double & returns the obtained relative precision via a reference.

`ifail`: int & returns an integer. Zero indicates success, if not zero the routine did not obtain the requested precision..

The function does a three dimensional integration over a hypercube. The underlying routine is `jbdadm1` which is a simple port to C++ of the CERNLIB[7] routine `DADMUL`. This in turn was based on [9].

Defined in `jbnulib.h`, implemented in `jbdadm1.cc`.

5 Chiral Perturbation Theory

The classic papers introducing ChPT are [10, 11, 12]. References to lectures and introductions can be found in [13]. A review at two-loop order is [14]. The notation used here correspond to the notation introduced by Gasser and Leutwyler, B, F, l_i^r, B_0 [11] and $F_0 L_i^r$ [12] for the two and three flavour case. In general the decay constants are defined with a normalization of $F_\pi \approx 92$ MeV. The coupling constants in the higher order Lagrangians are usually referred to as low-energy constants (LECs). Power counting is the usual dimensional counting with orders referred to as p^n with alternatively p^2 or lowest-order (LO), p^4 or next-to-leading-order (NLO) and p^6 or next-to-next-to-leading order (NNLO).

6 Data structures

This section describes a number of classes to deal with input parameters and LECs. The default value mechanism of C++ is used to give them initial values if not specified. These are visible below as “=value” in the definitions.

6.1 Three flavour ChPT

6.1.1 Class: physmass

`physmass(mpiin=0.135, mkin=0.495, metain=0.548, fpiin=0.0922, muin=0.77)`

`mpiin, mkin, metain, fpiin, muin: const double`

Private data: `double mpi, mk, meta, fpi, mu`

Physical quantities: pion, kaon and eta mass, pion-decay constant and subtraction scale μ .

Relevant physical case: three flavour ChPT, isospin limit.

Input member functions:

`void setmpi(const double mpiin=0.135)`

`void setmk(const double mkin=0.495)`

`void setmeta(const double metain=0.548)`

`void setfpi(const double fpiin=0.0922)`

`void setmu(const double muin=0.77)`

Output member functions exist in two varieties. Those that return all or a subset of values using references or those that return one value as the function value.

`void out(double &mpiout, double &mkout, double &metaout, double &fpiout,
double &muout)`

`double getmpi(void)`

`double getmk(void)`

`double getmeta(void)`

`double getfpi(void)`

`double getmu(void)`

Operators defined: `<<`, `>>` and `==`.

`<<` and `>>` are defined such that output and input streams work as expected. The input stream should be exactly in the format provided by the output stream.

`==` checks for equality within relative precision of 10^{-7} . An error will occur if any of the data members is zero.

Defined in `inputs.h`, implemented in `inputs.cc`, examples of use in `testinputs.cc`.

6.1.2 Class: lomass

`lomass(mp0in=0.135, mk0in=0.495, f0in=0.090, muin=0.77)`

`mp0in, mk0in, f0in, muin: const double`

`lomass(const quarkmass mass)`

Private data: `double mp0, mk0, f0, mu`

Physical quantities: lowest order pion mass, lowest order kaon mass, lowest order pion-decay constant and subtraction scale μ .

Relevant physical case: three flavour ChPT, isospin limit.

The constructor from a `quarkmass` is provided such that conversions can be used.

Input member functions:

```
void setmp0(const double mp0in=0.135) void setmk0(const double mk0in=0.495)
void setf0(const double f0in=0.09) void setmu(const double muin=0.77)
```

Output member functions exist in two varieties. Those that return all or a subset of values using references or those that return one value as the function value.

```
void out(double &mp0out, double &mk0out, double &f0out, double &muout)
double getmp0(void)
double getmk0(void)
double getf0(void)
double getmu(void)
```

Operators defined: `<<`, `>>` and `==`.

`<<` and `>>` are defined such that output and input streams work as expected. The input stream should be exactly in the format provided by the output stream.

`==` checks for equality within relative precision of 10^{-7} . An error will occur if any of the data members is zero.

Defined in `inputs.h`, implemented in `inputs.cc`, examples of use in `testinputs.cc`.

6.1.3 Class: `quarkmass`

```
quarkmass(B0mhatin=0.01, B0msin=0.25, f0in=0.090, muin=0.77)
```

```
B0mhatin,B0msin,f0in,muin: const double
```

```
quarkmass(const lomass mass)
```

```
Private data: double B0mhat,B0ms,f0,mu
```

Physical quantities: $B_0\hat{m}$, B_0m_s , lowest order pion-decay constant and subtraction scale μ .

The quantities $B_0m\hat{m}$ and B_0m_s are the LEC B_0 [12] multiplied by the up-down quark mass and strange quark mass respectively. These are independent of the QCD scale. The lowest order pion and kaon masses are given by $m_{\pi LO} = \sqrt{2B_0\hat{m}}$ and $m_{K LO} = \sqrt{B_0(\hat{m} + m_s)}$

Relevant physical case: three flavour ChPT, isospin limit.

The constructor from a `lomass` is provided such that conversions can be used.

Input member functions:

```
void setB0mhat(const double B0mhatin=0.01)
void setB0ms(const double B0msin=0.25)
void setf0(const double f0in=0.09)
void setmu(const double muin=0.77)
```

Output member functions exist in two varieties. Those that return all or a subset of values using references or those that return one value as the function value.

```

void out(double &B0mhatout, double &B0msout, double &f0out, double &muout)
double getB0mhat(void)
double getB0ms(void)
double getf0(void)
double getmu(void)

```

Operators defined: <<, >> and ==.

<< and >> are defined such that output and input streams work as expected. The input stream should be exactly in the format provided by the output stream.

== checks for equality within relative precision of 10^{-7} . An error will occur if any of the data members is zero.

Defined in `inputs.h`, implemented in `inputs.cc`, examples of use in `testinputs.cc`.

6.1.4 NLO LECs: Class Li

```

Li(l1r=0.,l2r=0.,l3r=0.,l4r=0.,l5r=0.,l6r=0.,l7r=0.,l8r=0.,l9r=0.,l10r=0.,
   h1r=0.,h2r=0.,mu=0.77,Name="nameless Li")

```

```
const double: l1r,...,l10r,h1r,h2r,mu
```

```
const string: Name
```

Private data: double L1r,L2r,L3r,L4r,L5r,L6r,L7r,L8r,L9r,L10r,H1r,H2r,mu and string name

Physical quantities the 12 LECs, L_i^r, H_i^r (of which two are so-called contact terms) of three-flavour ChPT as introduced in [12] and the subtraction scale μ .

Relevant physical case: three flavour ChPT

Input member functions:

```
void setli(const int n, const double lin)
```

```
void setli(const double lin, const int n)
```

Set the value of the LECs with index n . $n = 11, 12$ correspond to H_1^r, H_2^r .

```
setmu(const double muin)
```

Sets the scale μ to the value `muin`. This does *not* change the LECs, for that use `changescale`.

```
setname(const string namein)
```

Sets the name of the set of LECs.

Output member functions:

```
double out(const int n)
```

returns the value of the n 'th LEC.

`void out` exists in many varieties, 13 double references and a string returning all private data, 13 double references returning all LECs and the subtraction scale, 12 double references returning all LECs, 11 double references returning $L_1^r, \dots, L_{10}^r$ and the subtraction scale and 10 double references returning $L_1^r, \dots, L_{10}^r$.

```
void changescale(const double newmu)
```

This changes the subtraction scale to the new value given by `muin` and changes the LECs

according to the running derived in [12].

Operators defined: `<<`, `>>`, `+`, `-` and `*`.

`<<` and `>>` are defined such that output and input streams work as expected. The input stream should be exactly in the format provided by the output stream.

`*` allows to multiply an `Li` by a `double` in either order. The resulting value has all LECs multiplied by the value of the `double`.

`+` and `-` allow to add or subtract set of LECs. The resulting value of all LECs is the sum respectively the difference. A warning is printed if the scales are different.

Extra functions:

`Li Lirandom(void)`

`Li LirandomlargeNc(void)`

`Li LirandomlargeNc2(void)`

These return a set of random NLO LECs. The values are uniformly distributed between $\pm 1/(16\pi^2)$ for `Lirandom`. `LirandomlargeNc` does the same except that it leaves L_4^r , L_6^r and L_7^r zero. `LirandomlargeNc2` does the same but L_4^r , L_6^r and L_7^r get a random value between $\pm(1/3)/(16\pi^2)$. The random numbers are generated using the system generator `rand()` so initializing using something like `srand(time(0))`. These latter functions were used in the random walks in the L_i^r in [15].

Defined in `Li.h`, implemented in `Li.cc`, examples of use in `testLi.cc`.

In the subdirectory `test` there is a file `LiCiBE14.dat` that contains the last fit of the LECs [16].

6.1.5 NLO LECs: Class `Ci`

`Ci(Cr, mu=0.77, Name="nameless Ci")`

`Ci(mu=0.77, Name="nameless Ci")`

`const double: mu`

`const string: Name`

Private data: `double Cr[95]`, `mu` and `string name`

Physical quantities the 94 LECs, C_i^r (of which four are so-called contact terms) of three-flavour ChPT as introduced in [17, 18] and the subtraction scale μ . The C_i^r are the dimensionless version. Scale to the dimensionfull version with appropriate powers of F_0 but in practice normally with F_π .

Relevant physical case: three flavour ChPT

Input member functions:

`void setci(const int n, const double lin)`

`void setci(const double lin, const int n)`

Set the value of the LECs with index n .

`setmu(const double muin)`

Sets the scale μ to the value `muin`. This does *not* change the LECs, for that use `changescale`.

`setname(const string namein)` Sets the name of the set of LECs.

Output member functions:

`double out(const int n)` returns the value of the n 'th LEC.

`void out` exists in many varieties, with a `double Cit[95]`, a double reference and a string returning all private data, a `double Cit[95]`, a double reference returning all LECS and the subtraction scale, and a `double Cit[95]` returning the LECs only.

`void changescale(const double newmu, Li & Liin)`

`void changescale(Li & Liin, const double newmu)`

This changes the subtraction scale to the new value given by `muin` and changes the LECs according to the running derived in [18]. Note that it changes the scale of the values of the NLO LECs L_i^r in `Liin` as well.

Operators defined: `<<`, `>>`, `+`, `-` and `*`.

`<<` and `>>` are defined such that output and input streams work as expected. The input stream should be exactly in the format provided by the output stream.

`*` allows to multiply a `Ci` by a `double` in either order. The resulting value has all LECs multiplied by the value of the `double`.

`+` and `-` allow to add or subtract set of LECs. The resulting value of all LECs is the sum respectively the difference. A warning is printed if the scales are different.

Extra functions:

`Ci Cirandom(void)`

`Ci CirandomlargeNc(void)`

`Ci CirandomlargeNc2(void)`

These return a set of random NLO LECs. The values are uniformly distributed between $\pm 1/(16\pi^2)^2$ for `Cirandom`. `CirandomlargeNc` does the same except that it leaves all LECs that are not single trace terms zero. `CirandomlargeNc2` does the same but the non-single-trace terms get a LEC with a random value between $\pm(1/3)/(16\pi^2)^2$. The random numbers are generated using the system generator `rand()` so initializing using something like `srand(time(0))`. These latter functions were used in the random walks in the C_i^r in [15].

Defined in `Ci.h`, implemented in `Ci.cc`, examples of use in `testCi.cc`.

7 Loop integrals

Loop integrals are done with dimensional regularization and we use the standard ChPT variant of $\overline{MS}$. At one-loop order it was defined in [11, 12]. The definition at two-loop order can be found in [18].

We define for subtraction purposes:

$$d = 4 - 2\epsilon, \quad C = \ln(4\pi) + 1 - \gamma, \quad \lambda_0 = \frac{1}{\epsilon} + C, \quad \lambda_1 = \lambda_0 + C, \quad \lambda_2 = \lambda_0^2 + C^2. \quad (10)$$

The d -dimensional Feynman integrals do not depend directly on the subtraction scale. However, renormalization will always introduce the correct dependence. We define the one-loop integrals multiplied by an extra factor of $\mu^{2\epsilon}$ and the two-loop integrals with an extra factor of $\mu^{4\epsilon}$. This introduces the μ dependence in the expressions given below. References are to places where the integrals are defined and/or the method used elaborated.

7.1 Tadpole or one-propagator integrals

These are defined by

$$A(n, m^2) = \frac{\mu^{4-d}}{i} \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 - m^2)^n},$$

$$A(m^2, \mu^2) = A(1, m^2), \quad B(m^2, \mu^2) = A(2, m^2), \quad C(m^2, \mu^2) = A(3, m^2). \quad (11)$$

The expansions in ϵ are given by, see e.g. [19],

$$\begin{aligned} A(m^2, \mu^2) &= \frac{\lambda_0 m^2}{16\pi^2} + \bar{A}(m^2, \mu^2) + \epsilon A^\epsilon(m^2, \mu^2) + \mathcal{O}(\epsilon^2), \\ B(m^2, \mu^2) &= \frac{\lambda_0}{16\pi^2} + \bar{B}(m^2, \mu^2) + \epsilon B^\epsilon(m^2, \mu^2) + \mathcal{O}(\epsilon^2), \\ C(m^2, \mu^2) &= \bar{C}(m^2, \mu^2) + \epsilon C^\epsilon(m^2, \mu^2) + \mathcal{O}(\epsilon^2). \end{aligned} \quad (12)$$

The $\mathcal{O}(\epsilon)$ terms are further expanded as

$$\begin{aligned} A^\epsilon(m^2, \mu^2) &= \frac{m^2}{16\pi^2} \left(\frac{1}{2} C^2 - C \log \frac{m^2}{\mu^2} \right) + \bar{A}^\epsilon(m^2, \mu^2), \\ B^\epsilon(m^2, \mu^2) &= \frac{1}{16\pi^2} \left(\frac{1}{2} C^2 - C \log \frac{m^2}{\mu^2} - C \right) + \bar{B}^\epsilon(m^2, \mu^2), \\ C^\epsilon(m^2, \mu^2) &= \frac{1}{16\pi^2} \left(-\frac{C}{2m^2} \right) + \bar{C}^\epsilon(m^2, \mu^2). \end{aligned} \quad (13)$$

The analytical expressions are

$$\begin{aligned} \bar{A}(m^2, \mu^2) &= \frac{-m^2}{16\pi^2} \log \frac{m^2}{\mu^2} & \bar{A}^\epsilon(m^2, \mu^2) &= \frac{m^2}{16\pi^2} \left(\frac{1}{2} + \frac{\pi^2}{12} + \frac{1}{2} \log^2 \frac{m^2}{\mu^2} \right), \\ \bar{B}(m^2, \mu^2) &= \frac{1}{16\pi^2} \left(-1 - \log \frac{m^2}{\mu^2} \right) & \bar{B}^\epsilon(m^2, \mu^2) &= \frac{1}{16\pi^2} \left(\frac{1}{2} + \frac{\pi^2}{12} + \frac{1}{2} \log^2 \frac{m^2}{\mu^2} + \log \frac{m^2}{\mu^2} \right), \\ \bar{C}(m^2, \mu^2) &= \frac{1}{16\pi^2} \frac{-1}{2m^2} & \bar{C}^\epsilon(m^2, \mu^2) &= \frac{1}{16\pi^2} \left(\frac{1}{2m^2} + \frac{1}{2m^2} \log \frac{m^2}{\mu^2} \right), \end{aligned} \quad (14)$$

double Ab(const double msq, const double mu2): returns $\bar{A}(m^2, \mu^2)$.
double Bb(const double msq, const double mu2): returns $\bar{B}(m^2, \mu^2)$.
double Cb(const double msq, const double mu2): returns $\bar{C}(m^2, \mu^2)$.

double Abeps(const double msq, const double mu2): returns $\overline{A}^\epsilon(m^2, \mu^2)$.
double Bbeps(const double msq, const double mu2): returns $\overline{B}^\epsilon(m^2, \mu^2)$.
double Cbeps(const double msq, const double mu2): returns $\overline{C}^\epsilon(m^2, \mu^2)$.
double Ab(const int n, const double msq, const double mu2): returns $\overline{A}(m^2, \mu^2)$,
 $\overline{B}(m^2, \mu^2)$, $\overline{C}(m^2, \mu^2)$ for $n = 1, 2, 3$.

Defined in `oneloopintegrals.h`, implemented in `oneloopintegrals.cc`, examples of use in `testoneloopintegrals.cc`.

7.2 Bubbles or two-propagator integrals

7.2.1 Definitions

We first define the abbreviation

$$\langle X \rangle = \frac{\mu^{4-d}}{i} \int \frac{d^d q}{(2\pi)^d} \frac{X}{(q^2 - m_1^2)((q-p)^2 - m_2^2)}. \quad (15)$$

The bubble integrals themselves are defined by, see e.g. [20],

$$\begin{aligned} B(m_1^2, m_2^2, p^2, \mu^2) &= \langle 1 \rangle, \\ B_\mu(m_1^2, m_2^2, p, \mu^2) &= \langle q_\mu \rangle = p_\mu B_1(m_1^2, m_2^2, p^2, \mu^2), \\ B_{\mu\nu}(m_1^2, m_2^2, p, \mu^2) &= \langle q_\mu q_\nu \rangle = p_\mu p_\nu B_{21}(m_1^2, m_2^2, p^2, \mu^2) + g_{\mu\nu} B_{22}(m_1^2, m_2^2, p^2, \mu^2), \\ B_{\mu\nu\rho}(m_1^2, m_2^2, p, \mu^2) &= \langle q_\mu q_\nu q_\rho \rangle = p_\mu p_\nu p_\rho B_{31}(m_1^2, m_2^2, p^2, \mu^2) \\ &\quad + (g_{\mu\nu} p_\rho + g_{\mu\rho} p_\nu + g_{\rho\nu} p_\mu) B_{32}(m_1^2, m_2^2, p^2, \mu^2). \end{aligned} \quad (16)$$

The methods of [21] can be used to deduce the relations

$$\begin{aligned} B_1(m_1^2, m_2^2, p^2, \mu^2) &= -\frac{1}{2p^2} (A(m_1^2, \mu^2) - A(m_2^2, \mu^2) + (m_2^2 - m_1^2 - p^2) B(m_1^2, m_2^2, p^2, \mu^2)), \\ B_{22}(m_1^2, m_2^2, p^2, \mu^2) &= \frac{1}{2(d-1)} (A(m_2^2, \mu^2) + 2m_1^2 B(m_1^2, m_2^2, p^2, \mu^2) \\ &\quad + (m_2^2 - m_1^2 - p^2) B_1(m_1^2, m_2^2, p^2, \mu^2)), \\ B_{21}(m_1^2, m_2^2, p^2, \mu^2) &= \frac{1}{p^2} (A(m_2^2, \mu^2) + m_1^2 B(m_1^2, m_2^2, p^2, \mu^2) - dB_{22}(m_1^2, m_2^2, p^2, \mu^2)). \end{aligned} \quad (17)$$

This allows to rewrite all in terms of $B(m_1^2, m_2^2, p^2, \mu^2)$. These relations are used for the analytical evaluations given below.

The final evaluation is done by using a Feynman parameter x to combine the propagators and use the results for the tadpoles. The x integral needed can be done analytically or numerically.

The functions are then all expanded in terms of ϵ . The arguments of the various Bubble integrals are not written out.

$$B = \frac{\lambda_0}{16\pi^2} + \overline{B} + \epsilon B^\epsilon + \mathcal{O}(\epsilon^2),$$

$$\begin{aligned}
B_1 &= \frac{\lambda_0}{16\pi^2} \frac{1}{2} + \bar{B}_1 + \epsilon B_1^\epsilon + \mathcal{O}(\epsilon^2), \\
B_{21} &= \frac{\lambda_0}{16\pi^2} \frac{1}{3} + \bar{B}_{21} + \epsilon B_{21}^\epsilon + \mathcal{O}(\epsilon^2), \\
B_{22} &= \frac{\lambda_0}{16\pi^2} \left(\frac{m_1^2}{4} + \frac{m_2^2}{4} - \frac{p^2}{12} \right) + \bar{B}_{22} + \epsilon B_{22}^\epsilon + \mathcal{O}(\epsilon^2), \\
B_{31} &= \frac{\lambda_0}{16\pi^2} \frac{1}{4} + \bar{B}_{31} + \epsilon B_{31}^\epsilon + \mathcal{O}(\epsilon^2), \\
B_{32} &= \frac{\lambda_0}{16\pi^2} \left(\frac{m_1^2}{12} + \frac{m_2^2}{6} - \frac{p^2}{24} \right) + \bar{B}_{32} + \epsilon B_{32}^\epsilon + \mathcal{O}(\epsilon^2),
\end{aligned} \tag{18}$$

7.2.2 Analytical implementation

The functions in this section are all implemented fully analytically.

`const double: msq, m1sq, m2sq, psq, mu2` these are $m^2, m_1^2, m_2^2, p^2, \mu^2$.
`dcomplex Bb(m1sq, m2sq, psq, mu2)`: returns $\bar{B}(m_1^2, m_2^2, p^2, \mu^2)$
`dcomplex Bb(msq, psq, mu2)`: returns $\bar{B}(m^2, m^2, p^2, \mu^2)$ using the simpler equal mass formula.
`dcomplex B1b(m1sq, m2sq, psq, mu2)`: returns $\bar{B}_1(m_1^2, m_2^2, p^2, \mu^2)$
`dcomplex B21b(m1sq, m2sq, psq, mu2)`: returns $\bar{B}_{21}(m_1^2, m_2^2, p^2, \mu^2)$
`dcomplex B22b(m1sq, m2sq, psq, mu2)`: returns $\bar{B}_{22}(m_1^2, m_2^2, p^2, \mu^2)$
`dcomplex B22b(msq, psq, mu2)`: returns $\bar{B}_{22}(m^2, m^2, p^2, \mu^2)$ using the simpler equal mass formula.

Defined in `oneloopintegrals.h`, implemented in `oneloopintegrals.cc`, examples of use in `testoneloopintegrals.cc`.

7.2.3 Numerical implementation

The functions in this section are all implemented using a numerical complex integration over x . The integration routine used can be specified using the macro `WINTEGRAL` which defaults to `jbwgauss`. Any of the complex integration routines of `jbnumlib` can be used instead.

`const double: m1sq, m2sq, psq, mu2` these are m_1^2, m_2^2, p^2, μ^2 .
`dcomplex Bbnum(m1sq, m2sq, psq, mu2)`: returns $\bar{B}(m_1^2, m_2^2, p^2, \mu^2)$
`dcomplex B1bnum(m1sq, m2sq, psq, mu2)`: returns $\bar{B}_1(m_1^2, m_2^2, p^2, \mu^2)$
`dcomplex B21bnum(m1sq, m2sq, psq, mu2)`: returns $\bar{B}_{21}(m_1^2, m_2^2, p^2, \mu^2)$
`dcomplex B22bnum(m1sq, m2sq, psq, mu2)`: returns $\bar{B}_{22}(m_1^2, m_2^2, p^2, \mu^2)$
`dcomplex B31bnum(m1sq, m2sq, psq, mu2)`: returns $\bar{B}_{31}(m_1^2, m_2^2, p^2, \mu^2)$
`dcomplex B32bnum(m1sq, m2sq, psq, mu2)`: returns $\bar{B}_{32}(m_1^2, m_2^2, p^2, \mu^2)$

The precision of the numerical integration can be set and obtained:

`void setprecisiononeloopintegrals(const double eps)` sets the precision to `eps`.
`double getprecisiononeloopintegrals(void)` returns the present precision. The default is 1e-10.

Defined in `oneloopintegrals.h`, implemented in `oneloopintegrals.cc`, examples of use in `testoneloopintegrals.cc`:

7.3 Sunset integrals

7.3.1 Definition

We first define the abbreviation

$$\langle\langle X \rangle\rangle = \left(\frac{\mu^{4-d}}{i} \right) \int \frac{d^d r}{(2\pi)^d} \frac{d^d s}{(2\pi)^d} \frac{X}{(r^2 - m_1^2)(s^2 - m_2^2)((r+s-p)^2 - m_3^2)}. \quad (19)$$

The sunset integrals themselves are defined by

$$\begin{aligned} H(m_1^2, m_2^2, m_3^2, p^2, \mu^2) &= \langle\langle 1 \rangle\rangle, \\ H_\mu(m_1^2, m_2^2, m_3^2, p, \mu^2) &= \langle\langle r_\mu \rangle\rangle = p_\mu H_1(m_1^2, m_2^2, m_3^2, p^2, \mu^2), \\ H_{\mu\nu}(m_1^2, m_2^2, m_3^2, p, \mu^2) &= \langle\langle r_\mu r_\nu \rangle\rangle = p_\mu p_\nu H_{21}(m_1^2, m_2^2, m_3^2, p^2, \mu^2) \\ &\quad + g_{\mu\nu} H_{22}(m_1^2, m_2^2, m_3^2, p^2, \mu^2), \\ H_{\mu\nu\rho}(m_1^2, m_2^2, m_3^2, p, \mu^2) &= \langle\langle r_\mu r_\nu r_\rho \rangle\rangle = p_\mu p_\nu p_\rho H_{31}(m_1^2, m_2^2, m_3^2, p^2, \mu^2) \\ &\quad + (g_{\mu\nu} p_\rho + g_{\mu\rho} p_\nu + g_{\rho\nu} p_\mu) H_{32}(m_1^2, m_2^2, m_3^2, p^2, \mu^2). \end{aligned} \quad (20)$$

The needed integrals with s_μ replacing some of the r_μ in the definitions can be related to those without s_μ as described in [19]. The evaluation of these sunset integrals has been done in [19]. Further references can be found there.

We extract the parts the divergent parts and the parts containing C via

$$\begin{aligned} H(m_1^2, m_2^2, m_3^2, p^2, \mu^2) &= \frac{1}{(16\pi^2)^2} \left[(\lambda_2/2) (m_1^2 + m_2^2 + m_3^2) + (\lambda_1/2) (m_1^2 (1 - \log(m_1^2/\mu^2)) \right. \\ &\quad \left. + m_2^2 (1 - \log(m_2^2/\mu^2)) + m_3^2 (1 - \log(m_3^2/\mu^2)) - (p^2/2)) \right] \\ &\quad + H^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2) + \mathcal{O}(\epsilon), \end{aligned} \quad (21)$$

$$\begin{aligned} H_1(m_1^2, m_2^2, m_3^2, p^2, \mu^2) &= \frac{1}{(16\pi^2)^2} \left[(\lambda_2/4) (m_2^2 + m_3^2) + (\lambda_1/8) (2m_1^2 \right. \\ &\quad \left. + m_2^2 (1 - 4\log(m_2^2/\mu^2)) + m_3^2 (1 - 4\log(m_3^2/\mu^2)) - (2p^2/3)) \right] \\ &\quad + H_1^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2) + \mathcal{O}(\epsilon), \end{aligned} \quad (22)$$

$$H_{21}(m_1^2, m_2^2, m_3^2, p^2, \mu^2) = \frac{1}{(16\pi^2)^2} \left[(\lambda_2/6) (m_2^2 + m_3^2) + (\lambda_1/36) (3m_1^2 \right.$$

$$+ m_2^2(2 - 12 \log(m_2^2/\mu^2)) + m_3^2(2 - 12 \log(m_3^2/\mu^2)) - (3p^2/2)) \Big] + H_{21}^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2) + \mathcal{O}(\epsilon), \quad (23)$$

$$H_{31}(m_1^2, m_2^2, m_3^2, p^2, \mu^2) = \frac{1}{(16\pi^2)^2} \Big[(\lambda_2/8) (m_2^2 + m_3^2) + (\lambda_1/96) (4m_1^2 + m_2^2(3 - 24 \log(m_2^2/\mu^2)) + m_3^2(3 - 24 \log(m_3^2/\mu^2)) - (12p^2/5)) \Big] + H_{31}^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2) + \mathcal{O}(\epsilon), \quad (24)$$

The routines for the sunset integrals calculate the value at $p^2 = 0$ and the derivative there analytically. The remainder is then calculated with a rather smoot integral valid below threshold for the `double hh` functions and with a dispersive method for the `dcomplex zhh` functions. The latter is valid above and below threshold. The functions returning the derivative w.r.t. p^2 calculate the value at $p^2 = 0$ analytically and the remainder via a numerical integration as above.

7.3.2 Functions

The integration routines needed can be set using the macro `DINTEGRAL` for the real integration, default is `jbdgauss`, and `SINTEGRAL` for the real integration with a singularity, default is `jbdcauch`. Any of the similar routines in `jbnlib` can be used instead.

`const double: m1sq, m2sq, m3sq, psq, mu2:` these are $m_1^2, m_2^2, m_3^2, p^2, \mu^2$.

Valid below threshold:

`double hh(m1sq, m2sq, m3sq, psq, mu2)` returns $H^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hh1(m1sq, m2sq, m3sq, psq, mu2)` returns $H_1^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hh21(m1sq, m2sq, m3sq, psq, mu2)` returns $H_{21}^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hh31(m1sq, m2sq, m3sq, psq, mu2)` returns $H_{31}^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hhd(m1sq, m2sq, m3sq, psq, mu2)` returns $(\partial/\partial p^2)H^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hh1d(m1sq, m2sq, m3sq, psq, mu2)` returns $(\partial/\partial p^2)H_1^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hh21d(m1sq, m2sq, m3sq, psq, mu2)` returns $(\partial/\partial p^2)H_{21}^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$

Valid above and below threshold:

`dcomplex zhh(m1sq, m2sq, m3sq, psq, mu2)` returns $H^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`dcomplex zhh1(m1sq, m2sq, m3sq, psq, mu2)` returns $H_1^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`dcomplex zhh21(m1sq, m2sq, m3sq, psq, mu2)` returns $H_{21}^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`dcomplex zhh31(m1sq, m2sq, m3sq, psq, mu2)` returns $H_{31}^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`dcomplex zhhd(m1sq, m2sq, m3sq, psq, mu2)` returns $(\partial/\partial p^2)H^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`dcomplex zhh1d(m1sq, m2sq, m3sq, psq, mu2)` returns $(\partial/\partial p^2)H_1^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`dcomplex zhh21d(m1sq, m2sq, m3sq, psq, mu2)` returns $(\partial/\partial p^2)H_{21}^F(m_1^2, m_2^2, m_3^2, p^2, \mu^2)$

`void setprecisionsunsetintegrals(const double eps)` sets the precision to `eps`.

n	1	2	3	4	5	6	7	8
i	1	2	1	1	2	2	1	2
j	1	1	2	1	2	1	2	2
k	1	1	1	2	1	2	2	2

Table 1: The relation between the value of n and the powers i, j, k of the three propagators.

`double getprecisionsunsetintegrals(void)` returns the present precision. The default is `1e-10`.

Defined in `unsetintegrals.h`, implemented in `unsetintegrals.cc`, examples of use in `testunsetintegrals.cc`:

7.4 Sunsetintegrals with different powers of propagators

7.4.1 Definition

We first define the abbreviation

$$\langle\langle X \rangle\rangle_n = \left(\frac{\mu^{4-d}}{i} \right) \int \frac{d^d r}{(2\pi)^d} \frac{d^d s}{(2\pi)^d} \frac{X}{(r^2 - m_1^2)^i (s^2 - m_2^2)^j ((r+s-p)^2 - m_3^2)^k}. \quad (25)$$

The translation of n to values for i, j, k is given in Tab. 1. The sunset integrals themselves are defined by

$$\begin{aligned} H(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2) &= \langle\langle 1 \rangle\rangle_n, \\ H_\mu(n, m_1^2, m_2^2, m_3^2, p, \mu^2) &= \langle\langle r_\mu \rangle\rangle_n = p_\mu H_1(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2), \\ H_{\mu\nu}(n, m_1^2, m_2^2, m_3^2, p, \mu^2) &= \langle\langle r_\mu r_\nu \rangle\rangle_n = p_\mu p_\nu H_{21}(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2) \\ &\quad + g_{\mu\nu} H_{22}(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2). \end{aligned} \quad (26)$$

The needed integrals with s_μ replacing some of the r_μ in the definitions can be related to those without s_μ as described in [22, 23, 24, 25]. The evaluation of these sunsetintegrals is by the generalization of the methods of [19]. Further references can be found there.

The divergent parts and the parts containing C via taking derivatives w.r.t. masses of (21). We thus define the functions $H_i^F(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2)$ for all cases above, $i = 0(\text{blank}), 1, 21$.

The routines for the sunset integrals calculate the value at $p^2 = 0$ and the derivative there analytically. The remainder is then calculated with a rather smooth integral valid below threshold for the `double hh` functions. The functions returning the derivative w.r.t. p^2 calculate the value at $p^2 = 0$ analytically and the remainder via a numerical integration as above.

An added addition here is that case where the Kahlen function

$$\lambda(m_1^2, m_2^2, m_3^2) = \sqrt{(m_1^2 - m_2^2 - m_3^2)^2 - 4m_2^2 m_3^2}$$

vanishes, is treated correctly.

7.4.2 Functions

The integration routines needed can be set using the macro `DINTEGRAL` for the real integration, default is `jbdgauss`. Any of the similar routines in `jbnlib` can be used instead.

`const int n`: the integer n labelling the powers of the propagators as defined in Tab. 1.
`const double: m1sq, m2sq, m3sq, psq, mu2`: these are $m_1^2, m_2^2, m_3^2, p^2, \mu^2$.

Valid below threshold:

`double hh(n, m1sq, m2sq, m3sq, psq, mu2)` returns $H^F(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hh1(n, m1sq, m2sq, m3sq, psq, mu2)` returns $H_1^F(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hh21(n, m1sq, m2sq, m3sq, psq, mu2)` returns $H_{21}^F(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hhd(n, m1sq, m2sq, m3sq, psq, mu2)` returns $(\partial/\partial p^2)H^F(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hh1d(n, m1sq, m2sq, m3sq, psq, mu2)` returns $(\partial/\partial p^2)H_1^F(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2)$
`double hh21d(n, m1sq, m2sq, m3sq, psq, mu2)` returns $(\partial/\partial p^2)H_{21}^F(n, m_1^2, m_2^2, m_3^2, p^2, \mu^2)$

`void setprecisionquenchedsunsetintegrals(const double eps)` sets the precision to `eps`.

`double getprecisionquenchedsunsetintegrals(void)` returns the present precision. The default is `1e-10`.

Defined in `quenchedsunsetintegrals.h`, implemented in `quenchedsunsetintegrals.cc`, examples of use in `testquenchedsunsetintegrals.cc`:

7.5 Finite volume tadpole integrals

7.5.1 Definitions

The methods used for these are derived in detail in [8], references to earlier literature can be found there. The integrals used here are given in the Minkowski conventions as defined in [26]. All of the integrals are available with two different methods, one using a summation over Bessel function and the other an integral over a Jacobi theta function. The versions included at present are using periodic boundary conditions, all three spatial sizes of the same length L and the time direction of infinite extent.

The tadpole integrals A and $A_{\mu\nu}$ are defined as

$$\left\{ \tilde{A}^V(m^2, L, \mu^2), \tilde{A}_{\mu\nu}^V(m^2, L, \mu^2) \right\} = \frac{\mu^{4-d}}{i} \int_V \frac{d^d r}{(2\pi)^d} \frac{\{1, r_\mu r_\nu\}}{(r^2 - m^2)}. \quad (27)$$

The B tadpole integrals are the same but with a doubled propagator. The subscript V on the integral indicates that the integral is a discrete sum over the three spatial components and an integral over the remainder. The size of the spatial directions is L .

At finite volume, there are more Lorentz-structures possible. The tensor $t_{\mu\nu}$, the spatial part of the Minkowski metric $g_{\mu\nu}$, is needed for these. The functions for $\tilde{A}_{\mu\nu}^V$ are

$$\tilde{A}_{\mu\nu}^V(m^2, L, \mu^2) = g_{\mu\nu} \tilde{A}_{22}^V(m^2, L, \mu^2) + t_{\mu\nu} \tilde{A}_{23}^V(m^2, L, \mu^2). \quad (28)$$

In infinite volume A_{22} is related to A and A_{23} vanishes. The relation in finite volume is given by

$$d\tilde{A}_{22}^V(m^2, L, \mu^2) + 3\tilde{A}_{23}^V(m^2, L, \mu^2) = m^2 \tilde{A}^V(m^2, L, \mu^2). \quad (29)$$

The full integrals are now split in the infinite volume part which was defined earlier in Sect. 7.1 and the finite volume remainder as

$$\begin{aligned} \tilde{A}^V(m^2, L, \mu^2) &= \frac{\lambda_0 m^2}{16\pi^2} + \bar{A}(m^2, \mu^2) + \bar{A}^V(m^2, L) + \epsilon (A^\epsilon(m^2, \mu^2) + A^{V\epsilon}(m^2, L, \mu^2)) + \mathcal{O}(\epsilon^2), \\ \tilde{B}^V(m^2, L, \mu^2) &= \frac{\lambda_0}{16\pi^2} + \bar{B}(m^2, \mu^2) + \bar{B}^V(m^2, L) + \epsilon (B^\epsilon(m^2, \mu^2) + B^{V\epsilon}(m^2, L, \mu^2)) + \mathcal{O}(\epsilon^2), \\ \tilde{A}_{22}^V(m^2, L, \mu^2) &= \frac{\lambda_0 m^4}{4(16\pi^2)} + \bar{A}_{22}(m^2, \mu^2) + \bar{A}_{22}^V(m^2, L) + \epsilon (A_{22}^\epsilon(m^2, \mu^2) + A_{22}^{V\epsilon}(m^2, L, \mu^2)) \\ &\quad + \mathcal{O}(\epsilon^2), \\ \tilde{A}_{23}^V(m^2, L, \mu^2) &= \bar{A}_{23}^V(m^2, \mu^2) + \epsilon A_{22}^{V\epsilon}(m^2, L) + \mathcal{O}(\epsilon^2). \end{aligned} \quad (30)$$

7.5.2 Functions

The integration routines needed can be set using the macro `DINTEGRAL` for the real integration, default is `jbdgauss`. Any of the similar routines in `jbnunlib` can be used instead.

`const double: msq, L : msq` is m^2 and `L` is the size L of the spatial dimension.

Evaluated with theta functions:

`double AbVt(msq, L):` returns $\bar{A}^V(m^2, L)$.
`double BbVt(msq, L):` returns $\bar{B}^V(m^2, L)$.
`double A22bVt(msq, L):` returns $\bar{A}_{22}^V(m^2, L)$.
`double A23bVt(msq, L):` returns $\bar{A}_{23}^V(m^2, L)$.

Evaluated with Bessel functions:

`double AbVb(msq, L):` returns $\bar{A}^V(m^2, L)$.
`double BbVb(msq, L):` returns $\bar{B}^V(m^2, L)$.
`double A22bVb(msq, L):` returns $\bar{A}_{22}^V(m^2, L)$.
`double A23bVb(msq, L):` returns $\bar{A}_{23}^V(m^2, L)$.

The last letter indicates whether they are computed with the theta function or Bessel function method. The results were checked by comparing against each other and by comparing

with the independent Bessel function implementation done in [27].

`void setprecisionfinitevolumeoneloop(const double Abacc=1e-10, const double Bbacc=1e-9, const bool printout=true)` sets the precision for the finite volume integrals evaluated with theta function to `Abacc` for the tadpole integrals, `Bbacc` for the bubble integrals. The last variable `printout` is a logical variable which can be set to true or false, default is false. Default values are those indicated.

`void setprecisionfinitevolumeoneloopb(const int maxsum=100, const double Bbacc=1e-5, const bool printout=true)` sets the precision for the finite volume integrals evaluated with Bessel functions. The first argument indicates how far the sum over Bessel functions is taken. Maximum at present is 1000. The second argument gives the precision of the numerical integration for the bubble integrals. Defined in `finitevolumeoneloopintegrals.h`, implemented in `finitevolumeoneloopintegrals.cc`, examples of use in `testfinitevolumeoneloopintegrals.cc`.

7.6 Finite volume sunsetintegrals

7.6.1 Definitions

The sunset integrals are defined with

$$\langle\langle X \rangle\rangle_V = \frac{\mu^{8-2d}}{i^2} \int_V \frac{d^d r}{(2\pi)^d} \frac{d^d 1}{(2\pi)^d} \frac{\{1, r_\mu, r_\mu r_\nu\}}{(r^2 - m_1^2)(s^2 - m_2^2)((r+s-p)^2 - m_3^2)}. \quad (31)$$

The subscript V indicates that the spatial dimensions are a discrete sum rather than an integral. The conventions correspond to those in infinite volume of [19] and of Sect. 7.3. Integrals with the other momentum s in the numerator are related using the relations shown in [19] which remain valid at finite volume in the cms frame [8].

In the cms frame we define the functions¹

$$\begin{aligned} \tilde{H}_\mu^V &= \langle\langle X \rangle\rangle_V \\ \tilde{H}_\mu^V &= \langle\langle r_\mu \rangle\rangle_V = p_\mu \tilde{H}_1^V \\ \tilde{H}_{\mu\nu}^V &= \langle\langle r_\mu r_\nu \rangle\rangle_V = p_\mu p_\nu \tilde{H}_{21}^V + g_{\mu\nu} \tilde{H}_{22}^V + t_{\mu\nu} \tilde{H}_{27}^V. \end{aligned} \quad (32)$$

The arguments of all functions in the cms frame are $(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$. These functions satisfy in finite volume [8],

$$\begin{aligned} \tilde{H}_1^V + \tilde{H}_1^V(m_2^2, m_3^2, m_1^2, p^2, L, \mu^2) + \tilde{H}_1^V(m_3^2, m_1^2, m_2^2, p^2, L, \mu^2) &= \tilde{H}^V, \\ p^2 \tilde{H}_{21}^V + d \tilde{H}_{22}^V + 3 \tilde{H}_{27}^V - m_1^2 H &= \tilde{A}^V(m_2^2) \tilde{A}^V(m_3^2). \end{aligned} \quad (33)$$

¹In the cms frame $t_{\mu\nu} = g_{\mu\nu} - p_\mu p_\nu / p^2$ but the separation appears naturally in the calculation [8]. In addition, it avoids singularities in the limit $p \rightarrow 0$.

The arguments of the sunset functions in the relations, if not mentioned explicitly, are $(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$.

We split the functions in an infinite volume part, H_i , and a finite volume correction, H_i^V , with $\tilde{H}_i^V = H_i + H_i^V$. The infinite volume part has been discussed above. For the finite volume parts we define

$$\begin{aligned}
H^V &= \frac{\lambda_0}{16\pi^2} \left(\bar{A}^V(m_1^2) + \bar{A}^V(m_2^2) + \bar{A}^V(m_3^2) \right) + \frac{1}{16\pi^2} (A^{V\epsilon}(m_1^2) + A^{V\epsilon}(m_2^2) + A^{V\epsilon}(m_3^2)) \\
&\quad + H^{VF} + \mathcal{O}(\epsilon), \\
H_1^V &= \frac{\lambda_0}{16\pi^2} \frac{1}{2} \left(\bar{A}^V(m_2^2) + \bar{A}^V(m_3^2) \right) + \frac{1}{16\pi^2} \frac{1}{2} (A^{V\epsilon}(m_2^2) + A^{V\epsilon}(m_3^2)) + H_1^{VF} + \mathcal{O}(\epsilon), \\
H_{21}^V &= \frac{\lambda_0}{16\pi^2} \frac{1}{3} \left(\bar{A}^V(m_2^2) + \bar{A}^V(m_3^2) \right) + \frac{1}{16\pi^2} \frac{1}{3} (A^{V\epsilon}(m_2^2) + A^{V\epsilon}(m_3^2)) + H_{21}^{VF} + \mathcal{O}(\epsilon), \\
H_{27}^V &= \frac{\lambda_0}{16\pi^2} \left(\bar{A}_{23}^V(m_1^2) + \frac{1}{3} \bar{A}_{23}^V(m_2^2) + \frac{1}{3} \bar{A}_{23}^V(m_3^2) \right) \\
&\quad + \frac{1}{16\pi^2} \left(A_{23}^{V\epsilon}(m_1^2) + \frac{1}{3} A_{23}^{V\epsilon}(m_2^2) + \frac{1}{3} A_{23}^{V\epsilon}(m_3^2) \right) + H_{27}^{VF} + \mathcal{O}(\epsilon). \tag{34}
\end{aligned}$$

The finite parts are defined differently from the infinite volume case in [19]. The parts with $A^{V\epsilon}$ are removed here as well.

The functions H_i^{VF} can be computed with the methods of [8]. They are obtained by adding the parts labeled with G and H in Sect. 4.3 and the part of Sect. 4.4 in [8]. The derivatives w.r.t. p^2 can be treated using a simple adaptation of that method.

The method for evaluation works only below threshold. The numerical evaluation is rather slow. Playing with the precision settings for the specific case you need is very strongly recommended.

7.6.2 Functions

`const double:` `m1sq,m2sq,m3sq,psq,L,mu2`. These correspond to $m_1^2, m_2^2, m_3^2, p^2, L, \mu^2$.

Evaluation using theta functions:

```

double hhVt(m1sq,m2sq,m3sq,psq,L,mu2): returns  $H^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh1Vt(m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_1^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh21Vt(m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_{21}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh22Vt(m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_{22}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh27Vt(m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_{27}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hhdVt(m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh1dVt(m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_1^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh21dVt(m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{21}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh22dVt(m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{22}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh27dVt(m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{27}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .

```

Evaluation using Bessel functions:

```
double hhVb(m1sq,m2sq,m3sq,psq,L,mu2): returns  $H^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh1Vb(m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_1^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh21Vb(m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_{21}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh22Vb(m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_{22}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh27Vb(m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_{27}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hhdVb(m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh1dVb(m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_1^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh21dVb(m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{21}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh22dVb(m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{22}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh27dVb(m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{27}^{VF}(m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
```

For all cases discussed both methods, via Bessel or (generalized) Jacobi theta functions, give the same results. The derivatives w.r.t. p^2 for all the integrals were compared with taking a numerical derivative.

Note that the sunset functions at finite volume call the tadpole integrals evaluated with the same method. Do not forget to set precision for those as well.

```
void setprecisionfinitevolumesunsett(const double racc=1e-5,
    const double rsacc=1e-4, const bool printout=true)
```

The double values `sunsetracc` and `sunsetrsacc` set the accuracies of the numerical integration needed when one or two loop-momenta “feel” the finite volume. Default values are `1e-5` and `1e-4` respectively. The bool variable `printout` defaults to `true` and sets whether the setting is printed.

```
void setprecisionfinitevolumesunsetb(const int maxsum1=100,
    const int maxsum2=40, racc=1e-5, rsacc=1e-4, printout)
```

The integers `maxsum1` and `maxsum2` give how far the sum over Bessel functions is used for the case with one or two loop momenta “feeling” the finite volume. The first is maximum 1000, the second maximum 40 in the present implementation. In the latter case a triple sum is needed, hence the much lower upper bound. The double values `sunsetracc` and `sunsetrsacc` set the accuracies of the numerical integration which is still needed after the sum for both cases.

For most applications it makes sense to have a higher precision for the case with one loop momentum quantized, i.e. `racc` smaller than `rsacc`.

Defined in `finitevolumesunsetintegrals.h`, implemented in `finitevolumesunsetintegrals.cc` and examples of use in `testfinitevolumesunsetintegrals.cc`

7.7 Finite volume sunset integrals with different powers of propagators

7.7.1 Definitions

The sunset integrals with different powers of momenta are defined with

$$\langle\langle X \rangle\rangle_{nV} = \frac{\mu^{8-2d}}{i^2} \int_V \frac{d^d r}{(2\pi)^d} \frac{d^d 1}{(2\pi)^d} \frac{\{1, r_\mu, r_\mu r_\nu\}}{(r^2 - m_1^2)^i (s^2 - m_2^2)^j ((r + s - p)^2 - m_3^2)^k}. \quad (35)$$

The subscript V indicates that the spatial dimensions are a discrete sum rather than an integral. The translation of n to the powers i, j, k is given in Tab. 1.

In the cms frame we define the functions²

$$\begin{aligned} \tilde{H}_\mu^V &= \langle\langle X \rangle\rangle_{nV} \\ \tilde{H}_\mu^V &= \langle\langle r_\mu \rangle\rangle_{nV} = p_\mu \tilde{H}_1^V \\ \tilde{H}_{\mu\nu}^V &= \langle\langle r_\mu r_\nu \rangle\rangle_{nV} = p_\mu p_\nu \tilde{H}_{21}^V + g_{\mu\nu} \tilde{H}_{22}^V + t_{\mu\nu} \tilde{H}_{27}^V. \end{aligned} \quad (36)$$

The arguments of all functions in the cms frame are $(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$.

We split the functions in an infinite volume part, H_i , and a finite volume correction, H_i^V , with $\tilde{H}_i^V = H_i + H_i^V$. The infinite volume part has been discussed above. Note that the functions in the section are defined with the derivative w.r.t. m_l^2 for the propagators with mass m_l^2 . For the finite volume parts we define the subtraction as before in (34), i.e. we subtract the $A^{V\epsilon}, B^{V\epsilon}$ parts.

The functions H_i^{VF} can be computed with the methods of [8]. They are obtained by adding the parts labeled with G and H in Sect. 4.3 and the part of Sect. 4.4 in [8]. The derivatives w.r.t. p^2 can be treated using a simple adaptation of that method.

The method for evaluation works only below threshold. The numerical evaluation is rather slow. Playing with the precision settings for the specific case you need is very strongly recommended.

7.7.2 Functions

`const int n` The propagator cases as given in Tab. 1.

`const double: m1sq, m2sq, m3sq, psq, L, mu2.` These correspond to $m_1^2, m_2^2, m_3^2, p^2, L, \mu^2$.

Evaluation using theta functions:

`double hhVt(n, m1sq, m2sq, m3sq, psq, L, mu2):` returns $H^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$.

`double hh1Vt(n, m1sq, m2sq, m3sq, psq, L, mu2):` returns $H_1^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$.

`double hh21Vt(n, m1sq, m2sq, m3sq, psq, L, mu2):` returns $H_{21}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$.

`double hh22Vt(n, m1sq, m2sq, m3sq, psq, L, mu2):` returns $H_{22}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$.

²In the cms frame $t_{\mu\nu} = g_{\mu\nu} - p_\mu p_\nu / p^2$ but the separation appears naturally in the calculation [8]. In addition, it avoids singularities in the limit $p \rightarrow 0$.

```

double hh27Vt(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_{27}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hhdVt(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh1dVt(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_1^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh21dVt(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{21}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh22dVt(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{22}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh27dVt(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{27}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .

```

Evaluation using Bessel functions:

```

double hhVb(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $H^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh1Vb(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_1^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh21Vb(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_{21}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh22Vb(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_{22}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh27Vb(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $H_{27}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hhdVb(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh1dVb(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_1^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh21dVb(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{21}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh22dVb(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{22}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .
double hh27dVb(n,m1sq,m2sq,m3sq,psq,L,mu2): returns  $(\partial/\partial p^2)H_{27}^{VF}(n, m_1^2, m_2^2, m_3^2, p^2, L, \mu^2)$ .

```

For all cases discussed both methods, via Bessel or (generalized) Jacobi theta functions, give the same results. The derivatives w.r.t. p^2 for all the integrals were compared with taking a numerical derivative.

Note that the sunset functions at finite volume call the tadpole integrals evaluated with the same method. Do not forget to set precision for those as well.

```

void setprecisionfinitevolumesunsett(const double racc=1e-5,
    const double rsacc=1e-4,const bool printout=true)

```

The double values `sunsetracc` and `sunsetrsacc` set the accuracies of the numerical integration needed when one or two loop-momenta “feel” the finite volume. Default values are `1e-5` and `1e-4` respectively. The bool variable `printout` defaults to `true` and sets whether the setting is printed.

```

void setprecisionfinitevolumesunsetb(const int maxsum1=100,
    const int maxsum2=40,racc=1e-5,rsacc=1e-4,printout)

```

The integers `maxsum1` and `maxsum2` give how far the sum over Bessel functions is used for the case with one or two loop momenta “feeling” the finite volume. The first is maximum 1000, the second maximum 40 in the present implementation. In the latter case a triple sum is needed, hence the much lower upper bound. The double values `sunsetracc` and `sunsetrsacc` set the accuracies of the numerical integration which is still needed after the sum for both cases.

For most applications it makes sense to have a higher precision for the case with one loop

momentum quantized, i.e. `racc` smaller than `rsacc`.

Defined in `finitevolumesunsetintegrals.h`, implemented in `finitevolumesunsetintegrals.cc` and examples of use in `testfinitevolumesunsetintegrals.cc`

8 Three flavour isospin conserving results

8.1 Masses, decay constants and vacuum-expectation-values: in physical

The exansions in this subsection are defined in terms of the physical masses, m_π, m_K, m_η and the physical pion decay constant F_π .

8.1.1 Masses

The masses of the pion, kaon and eta at two-loops in three flavour ChPT were evaluated in [19]. The pion and eta mass were done earlier with a different subtraction scheme and a different way to perform the sunset integrals in [28].

The expressions for the physical masses for $a = \pi, K, \eta$ are given by

$$m_{a\text{phys}}^2 = m_{a0}^2 + m_a^{2(4)} + m_a^{2(6)}. \quad (37)$$

The superscripts indicate the order of the diagrams in p that each contribution comes from. The lowest order masses are

$$m_{\pi 0}^2 = 2B_0\hat{m}, \quad m_{K 0}^2 = B_0(\hat{m} + m_s), \quad m_{\eta 0}^2 = \frac{2}{3}(\hat{m} + 2m_2). \quad (38)$$

The higher order contributions are split in the parts depending on the NLO LECs L_i^r , on the NNLO LECs C_i^r and the remainder as

$$m_a^{2(4)} = m_{aL}^{2(4)} + m_{aR}^{2(4)}, \quad m_a^{2(6)} = m_{aL}^{2(6)} + m_{aC}^{2(6)} + m_{aR}^{2(6)}. \quad (39)$$

The expressions for these can be found in [19] and on [13]. Note that when combining these with results from other sources one should be sure to use a compatible LO and NLO.

Pion mass:

```
double mpi4(phymass,Li) returns  $m_\pi^{2(4)}$ 
double mpi4L(phymass,Li) returns  $m_{\pi L}^{2(4)}$ 
double mpi4R(phymass,Li) returns  $m_{\pi R}^{2(4)}$ 
double mpi6(phymass,Li,Ci) returns  $m_\pi^{2(6)}$ 
double mpi6L(phymass,Li) returns  $m_{\pi L}^{2(6)}$ 
double mpi6C(phymass,Ci) returns  $m_{\pi C}^{2(6)}$ 
```

`double mpi6R(phymass)` returns $m_{\pi R}^{2(6)}$

Kaon mass:

`double mk4(phymass, Li)` returns $m_K^{2(4)}$
`double mk4L(phymass, Li)` returns $m_{KL}^{2(4)}$
`double mk4R(phymass, Li)` returns $m_{KR}^{2(4)}$
`double mk6(phymass, Li, Ci)` returns $m_K^{2(6)}$
`double mk6L(phymass, Li)` returns $m_{KL}^{2(6)}$
`double mk6C(phymass, Ci)` returns $m_{KC}^{2(6)}$
`double mk6R(phymass)` returns $m_{KR}^{2(6)}$

Eta mass:

`double meta4(phymass, Li)` returns $m_\eta^{2(4)}$
`double meta4L(phymass, Li)` returns $m_{\eta L}^{2(4)}$
`double meta4R(phymass, Li)` returns $m_{\eta R}^{2(4)}$
`double meta6(phymass, Li, Ci)` returns $m_\eta^{2(6)}$
`double meta6L(phymass, Li)` returns $m_{\eta L}^{2(6)}$
`double meta6C(phymass, Ci)` returns $m_{\eta C}^{2(6)}$
`double meta6R(phymass)` returns $m_{\eta R}^{2(6)}$

The functions are defined in `massesdecayvev.h`, implemented in `massesdecayvev.cc` and examples of use are in `testmassdecayvev.cc`.

8.1.2 Decay constants

The decay constants of the pion, kaon and eta at two-loops in three flavour ChPT were obtained in [19]. The pion and eta decay constants were done earlier with a different subtraction scheme and a different way to perform the sunset integrals in [28].

The expressions for the decay constants for $a = \pi, K, \eta$ are given by

$$F_{a \text{ phys}} = F_0 \left(1 + F_a^{(4)} + F_a^{(6)} \right). \quad (40)$$

The superscripts indicate the order of the diagrams in p that each contribution comes from. F_0 denotes the decay constant in the three-flavour chiral limit. The expressions were originally derived in [19], but note the description in the erratum of [29]. The expressions corrected for the error can be found in the website [13]. The normalization is such that $F_\pi \approx 92$ MeV.

The contributions themselves are divided into the parts depending on the NLO LECs L_i^r , on the NNLO LECs C_i^r and the remainder as

$$F_a^{(4)} = F_{aL}^{(4)} + F_{aR}^{(4)}, \quad F_a^{(6)} = F_{aL}^{(6)} + F_{aC}^{(6)} + F_{aR}^{(6)}. \quad (41)$$

For the η the decay constant has been defined with the octet axial-vector current.

Pion decay constant:

```
double fpi4(phymass,Li) returns  $F_\pi^{(4)}$ 
double fpi4L(phymass,Li) returns  $F_{\pi L}^{(4)}$ 
double fpi4R(phymass,Li) returns  $F_{\pi R}^{(4)}$ 
double fpi6(phymass,Li,Ci) returns  $F_\pi^{(6)}$ 
double fpi6L(phymass,Li) returns  $F_{\pi L}^{(6)}$ 
double fpi6C(phymass,Ci) returns  $F_{\pi C}^{(6)}$ 
double fpi6R(phymass) returns  $F_{\pi R}^{(6)}$ 
```

Kaon decay constant:

```
double fk4(phymass,Li) returns  $F_K^{(4)}$ 
double fk4L(phymass,Li) returns  $F_{K L}^{(4)}$ 
double fk4R(phymass,Li) returns  $F_{K R}^{(4)}$ 
double fk6(phymass,Li,Ci) returns  $F_K^{(6)}$ 
double fk6L(phymass,Li) returns  $F_{K L}^{(6)}$ 
double fk6C(phymass,Ci) returns  $F_{K C}^{(6)}$ 
double fk6R(phymass) returns  $F_{K R}^{(6)}$ 
```

Eta decay constant:

```
double feta4(phymass,Li) returns  $F_\eta^{(4)}$ 
double feta4L(phymass,Li) returns  $F_{\eta L}^{(4)}$ 
double feta4R(phymass,Li) returns  $F_{\eta R}^{(4)}$ 
double feta6(phymass,Li,Ci) returns  $F_\eta^{(6)}$ 
double feta6L(phymass,Li) returns  $F_{\eta L}^{(6)}$ 
double feta6C(phymass,Ci) returns  $F_{\eta C}^{(6)}$ 
double feta6R(phymass) returns  $F_{\eta R}^{(6)}$ 
```

The functions are defined in `massesdecayvev.h`, implemented in `massesdecayvev.cc` and examples of use are in `testmassdecayvev.cc`.

8.1.3 getfpimeta

A problem that occurs in trying to compare to lattice QCD is that many of the routines are written in terms of the physical pion decay constant and physical masses. In particular, the eta mass is treated as physical. One thus needs a consistent eta mass and pion decay constant when varying the input pion and kaon mass. This assumes we have fitted the LECs L_i^r and C_i^r with a known set of $m_\pi, m_K, m_\eta, F_\pi$.

With that input we can obtain an eta mass and pion decay constant with as input values

the original `Liin`, `Ciin` and the `massin`. The formulas used are (39) and (41) up to order p^6 and p^4 . The solution is obtained by iteration and stops when six digits of precision are reached. This method was used in [26] to obtain the consistent set of masses and decay constants used there.

```
physmass getfpimeta6(const double mpiin, const double mkin,
    const physmass massin, const Li Liin, const Ci Ciin)
returns a physmass containing mpiin, mkin and the calculated compatibe meta, fpi with
the formulas including order  $p^6$ , i.e. to NNLO.
physmass getfpimeta4(const double mpiin, const double mkin,
    const physmass massin, const Li Liin)
returns a physmass containing mpiin, mkin and the calculated compatibe meta, fpi with
the formulas including order  $p^4$ , i.e. to NLO.
The functions are defined in getfpimeta.h, implemented in getfpimeta.cc and examples
of use are in testgetfpimeta.cc.
```

8.1.4 Vacuum-expectation-values

The corrections to the vacuum expectation values (vevs) $\langle 0|\bar{q}q|0\rangle$ for up, down and strange quarks in the isospin limit were calculated at two-loops in three flavour ChPT in [29]. The expression for the up and down quark vev are identical since we are in the isospin limit. We write the expressions in a form analogous to the decay constant treatment:

$$\langle 0|\bar{q}q|0\rangle_{a\text{phys}} = -F_0^2 B_0 \left(1 + \langle 0|\bar{q}q|0\rangle_a^{(4)} + \langle 0|\bar{q}q|0\rangle_a^{(6)} \right). \quad (42)$$

The superscripts indicate the order of the diagrams in p that each contribution comes from. The lowest order values are $-F_0^2 B_0$.

Note that the vevs are not directly measurable quantities. They depend on exactly the way the scalar densities are defined in QCD. ChPT can be used for them when a massin-dependent, chiral symmetry respecting subtraction scheme is used. $\overline{M}\overline{S}$ in QCD satisfies this, but there are other possibilities. Even within a scheme, B_0 and the quark masses depend on the QCD subtraction scale μ_{QCD} in such a way that $B_0 m_q$ is independent of it. The higher order corrections in this case also depend on the LECs for fully local counter-terms, H_1^r, H_2^r at order p^4 and $C_{91}^r, \dots, C_{94}^r$ at p^6 . When the scalar density is fully defined, measuring these quantities in e.g. lattice QCD and comparing with the ChPT expressions is a well defined procedure.

The contributions at the different orders themselves are split in the parts depending on the NLO LECs L_i^r , on the NNLO LECs C_i^r and the remainder as

$$\begin{aligned} \langle 0|\bar{q}q|0\rangle_a^{(4)} &= \langle 0|\bar{q}q|0\rangle_{aL}^{(4)} + \langle 0|\bar{q}q|0\rangle_{aR}^{(4)}, \\ \langle 0|\bar{q}q|0\rangle_a^{(6)} &= \langle 0|\bar{q}q|0\rangle_{aL}^{(6)} + \langle 0|\bar{q}q|0\rangle_{aC}^{(6)} + \langle 0|\bar{q}q|0\rangle_{aR}^{(6)}. \end{aligned} \quad (43)$$

These are defined for $q = u, s$.

$\langle 0|\bar{q}q|0\rangle_{u\text{ phys}}:$
`double qqup4(physmass, Li)` returns $\langle 0|\bar{q}q|0\rangle_u^{(4)}$
`double qqup4L(physmass, Li)` returns $\langle 0|\bar{q}q|0\rangle_{uL}^{(4)}$
`double qqup4R(physmass)` returns $\langle 0|\bar{q}q|0\rangle_{uR}^{(4)}$
`double qqup6(physmass, Li, Ci)` returns $\langle 0|\bar{q}q|0\rangle_u^{(6)}$
`double qqup6L(physmass, Li)` returns $\langle 0|\bar{q}q|0\rangle_{uL}^{(6)}$
`double qqup6C(physmass, Li)` returns $\langle 0|\bar{q}q|0\rangle_{uC}^{(6)}$
`double qqup6R(physmass)` returns $\langle 0|\bar{q}q|0\rangle_{uR}^{(6)}$

$\langle 0|\bar{q}q|0\rangle_{s\text{ phys}}:$
`double qqstrange4(physmass, Li)` returns $\langle 0|\bar{q}q|0\rangle_s^{(4)}$
`double qqstrange4L(physmass, Li)` returns $\langle 0|\bar{q}q|0\rangle_{sL}^{(4)}$
`double qqstrange4R(physmass)` returns $\langle 0|\bar{q}q|0\rangle_{sR}^{(4)}$
`double qqstrange6(physmass, Li, Ci)` returns $\langle 0|\bar{q}q|0\rangle_s^{(6)}$
`double qqstrange6L(physmass, Li)` returns $\langle 0|\bar{q}q|0\rangle_{sL}^{(6)}$
`double qqstrange6C(physmass, Li)` returns $\langle 0|\bar{q}q|0\rangle_{sC}^{(6)}$
`double qqstrange6R(physmass)` returns $\langle 0|\bar{q}q|0\rangle_{sR}^{(6)}$

The functions are defined in `massesdecayvev.h`, implemented in `massesdecayvev.cc` and examples of use are in `testmassesdecayvev.cc`

8.2 Masses, decay constants and vacuum-expectation-values: in lowest order

The exansions in this subsection are defined in terms of the lowest order masses, $m_{\pi 0}, m_{K 0}, m_{\eta} = \sqrt{(4m_{K 0}^2 - m_{\pi 0}^2)}/3$ and the lowest order, or chiral limit, pion decay constant F_0 .

8.2.1 Masses: in lowest order

The masses of the pion, kaon and eta at two-loops in three flavour ChPT were evaluated in [19]. The pion and eta mass were done earlier with a different subtraction scheme and a different way to perform the sunset integrals in [28].

The expressions for the physical masses for $a = \pi, K, \eta$ are given by

$$m_{a\text{ phys}}^2 = m_{a0}^2 + m_a^{2(4)0} + m_a^{2(6)0} . \quad (44)$$

The superscripts indicate the order of the diagrams in p that each contribution comes from and the extra 0 that it is defined in terms of lowest-order quantities.

The lowest order masses are

$$m_{\pi 0}^2 = 2B_0\hat{m} , \quad m_{K 0}^2 = B_0(\hat{m} + m_s) , \quad m_{\eta 0}^2 = \frac{2}{3}(\hat{m} + 2m_2) . \quad (45)$$

The higher order contributions are split in the parts depending on the NLO LECs L_i^r , on the NNLO LECs C_i^r and the remainder as

$$m_a^{2(4)0} = m_{aL}^{2(4)0} + m_{aR}^{2(4)0}, \quad m_a^{2(6)0} = m_{aL}^{2(6)0} + m_{aC}^{2(6)0} + m_{aR}^{2(6)0}. \quad (46)$$

The expressions for were derived during the work for [19] and on [13]. Note that when combining these with results from other sources one should be sure to use a compatible LO and NLO.

Pion mass:

```
double mpi4lo(lomass,Li) returns  $m_\pi^{2(4)0}$ 
double mpi4Llo(lomass,Li) returns  $m_{\pi L}^{2(4)0}$ 
double mpi4Rlo(lomass,Li) returns  $m_{\pi R}^{2(4)0}$ 
double mpi6lo(lomass,Li,Ci) returns  $m_\pi^{2(6)0}$ 
double mpi6Llo(lomass,Li) returns  $m_{\pi L}^{2(6)0}$ 
double mpi6Clo(lomass,Ci) returns  $m_{\pi C}^{2(6)0}$ 
double mpi6Rlo(lomass) returns  $m_{\pi R}^{2(6)0}$ 
```

Kaon mass:

```
double mk4lo(lomass,Li) returns  $m_K^{2(4)0}$ 
double mk4Llo(lomass,Li) returns  $m_{K L}^{2(4)0}$ 
double mk4Rlo(lomass,Li) returns  $m_{K R}^{2(4)0}$ 
double mk6lo(lomass,Li,Ci) returns  $m_K^{2(6)0}$ 
double mk6Llo(lomass,Li) returns  $m_{K L}^{2(6)0}$ 
double mk6Clo(lomass,Ci) returns  $m_{K C}^{2(6)0}$ 
double mk6Rlo(lomass) returns  $m_{K R}^{2(6)0}$ 
```

Eta mass:

```
double meta4lo(lomass,Li) returns  $m_\eta^{2(4)0}$ 
double meta4Llo(lomass,Li) returns  $m_{\eta L}^{2(4)0}$ 
double meta4Rlo(lomass,Li) returns  $m_{\eta R}^{2(4)0}$ 
double meta6lo(lomass,Li,Ci) returns  $m_\eta^{2(6)0}$ 
double meta6Llo(lomass,Li) returns  $m_{\eta L}^{2(6)0}$ 
double meta6Clo(lomass,Ci) returns  $m_{\eta C}^{2(6)0}$ 
double meta6Rlo(lomass) returns  $m_{\eta R}^{2(6)0}$ 
```

The functions are defined in `massesdecayvevlo.h`, implemented in `massesdecayvevlo.cc` and examples of use are in `testmassdecayvevlo.cc`.

8.2.2 Decay constants: in lowest order

The decay constants of the pion, kaon and eta at two-loops in three flavour ChPT were obtained in [19]. The pion and eta decay constants were done earlier with a different subtraction scheme and a different way to perform the sunset integrals in [28].

The expressions for the decay constants for $a = \pi, K, \eta$ are given by

$$F_{a\text{phys}} = F_0 (1 + F_a^{(4)0} + F_a^{(6)0}) . \quad (47)$$

The superscripts indicate the order of the diagrams in p that each contribution comes from. The extra 0 indicates that the expansion is in terms of lowest-order quantities. F_0 denotes the decay constant in the three-flavour chiral limit. The expressions were originally derived during the work for [19] and can be found in the website [13]. The normalization is such that $F_\pi \approx 92$ MeV.

The contributions themselves are divided into the parts depending on the NLO LECs L_i^r , on the NNLO LECs C_i^r and the remainder as

$$F_a^{(4)0} = F_{aL}^{(4)0} + F_{aR}^{(4)0} , \quad F_a^{(6)0} = F_{aL}^{(6)0} + F_{aC}^{(6)0} + F_{aR}^{(6)0} . \quad (48)$$

For the η the decay constant has been defined with the octet axial-vector current.

Pion decay constant:

```
double fpi4lo(lomass,Li) returns  $F_\pi^{(4)0}$ 
double fpi4Llo(lomass,Li) returns  $F_{\pi L}^{(4)0}$ 
double fpi4Rlo(lomass,Li) returns  $F_{\pi R}^{(4)0}$ 
double fpi6lo(lomass,Li,Ci) returns  $F_\pi^{(6)0}$ 
double fpi6Llo(lomass,Li) returns  $F_{\pi L}^{(6)0}$ 
double fpi6Clo(lomass,Ci) returns  $F_{\pi C}^{(6)0}$ 
double fpi6Rlo(lomass) returns  $F_{\pi R}^{(6)0}$ 
```

Kaon decay constant:

```
double fk4lo(lomass,Li) returns  $F_K^{(4)0}$ 
double fk4Llo(lomass,Li) returns  $F_{KL}^{(4)0}$ 
double fk4Rlo(lomass,Li) returns  $F_{KR}^{(4)0}$ 
double fk6lo(lomass,Li,Ci) returns  $F_K^{(6)0}$ 
double fk6Llo(lomass,Li) returns  $F_{KL}^{(6)0}$ 
double fk6Clo(lomass,Ci) returns  $F_{KC}^{(6)0}$ 
double fk6Rlo(lomass) returns  $F_{KR}^{(6)0}$ 
```

Eta decay constant:

```
double feta4lo(lomass,Li) returns  $F_\eta^{(4)0}$ 
double feta4Llo(lomass,Li) returns  $F_{\eta L}^{(4)0}$ 
double feta4Rlo(lomass,Li) returns  $F_{\eta R}^{(4)0}$ 
```

`double feta6lo(lomass,Li,Ci)` returns $F_\eta^{(6)0}$
`double feta6Llo(lomass,Li)` returns $F_{\eta L}^{(6)0}$
`double feta6Clo(lomass,Ci)` returns $F_{\eta C}^{(6)0}$
`double feta6Rlo(lomass)` returns $F_{\eta R}^{(6)0}$

The functions are defined in `massesdecayvevlo.h`, implemented in `massesdecayvevlo.cc` and examples of use are in `testmassdecayvevlo.cc`.

8.2.3 Vacuum-expectation-values: in lowest order

The corrections to the vacuum expectation values (vevs) $\langle 0|\bar{q}q|0\rangle$ for up, down and strange quarks in the isospin limit were calculated at two-loops in three flavour ChPT in [29]. The expression for the up and down quark vev are identical since we are in the isospin limit. We write the expressions in a form analogous to the decay constant treatment:

$$\langle 0|\bar{q}q|0\rangle_{a\text{phys}} = -F_0^2 B_0 \left(1 + \langle 0|\bar{q}q|0\rangle_a^{(4)0} + \langle 0|\bar{q}q|0\rangle_a^{(6)0} \right). \quad (49)$$

The superscripts indicate the order of the diagrams in p that each contribution comes from. The extra 0 indicates that the expansion is defined in terms of lowest-order quantities. The lowest order values are $-F_0^2 B_0$.

Note that the vevs are not directly measurable quantities. They depend on exactly the way the scalar densities are defined in QCD. ChPT can be used for them when a massindependent, chiral symmetry respecting subtraction scheme is used. $\overline{MS}$ in QCD satisfies this, but there are other possibilities. Even within a scheme, B_0 and the quark masses depend on the QCD subtraction scale μ_{QCD} in such a way that $B_0 m_q$ is independent of it. The higher order corrections in this case also depend on the LECs for fully local counterterms, H_1^r, H_2^r at order p^4 and $C_{91}^r, \dots, C_{94}^r$ at p^6 . When the scalar density is fully defined, measuring these quantities in e.g. lattice QCD and comparing with the ChPT expressions is a well defined procedure.

The contributions at the different orders themselves are split in the parts depending on the NLO LECs L_i^r , on the NNLO LECs C_i^r and the remainder as

$$\begin{aligned} \langle 0|\bar{q}q|0\rangle_a^{(4)0} &= \langle 0|\bar{q}q|0\rangle_{aL}^{(4)0} + \langle 0|\bar{q}q|0\rangle_{aR}^{(4)0}, \\ \langle 0|\bar{q}q|0\rangle_a^{(6)0} &= \langle 0|\bar{q}q|0\rangle_{aL}^{(6)0} + \langle 0|\bar{q}q|0\rangle_{aC}^{(6)0} + \langle 0|\bar{q}q|0\rangle_{aR}^{(6)0}. \end{aligned} \quad (50)$$

These are defined for $q = u, s$.

$\langle 0|\bar{q}q|0\rangle_{u\text{phys}}$:
`double qqup4lo(lomass,Li)` returns $\langle 0|\bar{q}q|0\rangle_u^{(4)0}$
`double qqup4Llo(lomass,Li)` returns $\langle 0|\bar{q}q|0\rangle_{uL}^{(4)0}$
`double qqup4Rlo(lomass)` returns $\langle 0|\bar{q}q|0\rangle_{uR}^{(4)0}$
`double qqup6lo(lomass,Li,Ci)` returns $\langle 0|\bar{q}q|0\rangle_u^{(6)0}$
`double qqup6Llo(lomass,Li)` returns $\langle 0|\bar{q}q|0\rangle_{uL}^{(6)0}$

```

double qqup6Clo(lomass,Li) returns  $\langle 0|\bar{q}q|0\rangle_{uC}^{(6)0}$ 
double qqup6Rlo(lomass) returns  $\langle 0|\bar{q}q|0\rangle_{uR}^{(6)0}$ 

 $\langle 0|\bar{q}q|0\rangle_{s\text{ phys}}$ :
double qqstrange4lo(lomass,Li) returns  $\langle 0|\bar{q}q|0\rangle_s^{(4)0}$ 
double qqstrange4Llo(lomass,Li) returns  $\langle 0|\bar{q}q|0\rangle_{sL}^{(4)0}$ 
double qqstrange4Rlo(lomass) returns  $\langle 0|\bar{q}q|0\rangle_{sR}^{(4)0}$ 
double qqstrange6lo(lomass,Li,Ci) returns  $\langle 0|\bar{q}q|0\rangle_s^{(6)0}$ 
double qqstrange6Llo(lomass,Li) returns  $\langle 0|\bar{q}q|0\rangle_{sL}^{(6)0}$ 
double qqstrange6Clo(lomass,Li) returns  $\langle 0|\bar{q}q|0\rangle_{sC}^{(6)0}$ 
double qqstrange6Rlo(lomass) returns  $\langle 0|\bar{q}q|0\rangle_{sR}^{(6)0}$ 

```

The functions are defined in `massesdecayvevlo.h`, implemented in `massesdecayvevlo.cc` and examples of use are in `testmassesdecayvevlo.cc`

8.3 Masses and decay constants at finite volume

The expressions treated in this section have been derived in [26]. A general remark is that care should be taken to set the precision in the loop integrals sufficiently high. For the one-loop integrals setting it very high is usually no problem. For the sunset integrals the evaluation can become very slow. It is strongly recommended to play around with the settings and compare the outputs for the two ways to evaluate the integral. The theta and Bessel function evaluation approach the correct answer differently. For most cases it is possible to have `rsacc` set smaller than `racc`.

For many applications it is useful to calculate the very time consuming parts, those labeled `6RV`, once and store them. They only depend nontrivially on the masses and size of the finite volume. The decay constant dependence is very simple, an overall factor at each order, and there is dependence on the NLO LECs L_i^r .

The results presented in this section are with periodic boundary conditions and an infinite extension in the time direction. They are also restricted to the case where the particle is at rest, i.e. $\vec{p} = 0$.

8.3.1 Masses at finite volume

The finite volume corrections to the masses squared³ are defined as the difference of the mass squared in finite volume and in infinite volume:

$$\begin{aligned}
\Delta^V m_a^2 &= m_a^{2V} - m_a^{2V=\infty} = m_a^{2V(4)} + m_a^{2V(6)} . \\
m_a^{2V(6)} &= m_{aL}^{2V(6)} + m_{aR}^{2V(6)} .
\end{aligned} \tag{51}$$

These definitions are for $a = \pi, K, \eta$.

³Note that in other papers the corrections to the mass itself are sometimes quoted.

Pion mass (theta function method):

```
double mpi4Vt(const physmass massin,const double L) returns  $m_{\pi}^{2V(4)}$  .  
double mpi6Vt(const physmass massin,const Li Liin,const double L) returns  $m_{\pi}^{2V(6)}$  .  
double mpi6VLt(const physmass massin,const Li Liin,const double L) returns  $m_{\pi L}^{2V(6)}$  .  
double mpi6VRt(const physmass massin,const double L) returns  $m_{\pi R}^{2V(6)}$  .
```

Pion mass (Bessel function method):

```
double mpi4Vb(const physmass massin,const double L) returns  $m_{\pi}^{2V(4)}$  .  
double mpi6Vb(const physmass massin,const Li Liin,const double L) returns  $m_{\pi}^{2V(6)}$  .  
double mpi6VLb(const physmass massin,const Li Liin,const double L) returns  $m_{\pi L}^{2V(6)}$  .  
double mpi6VRb(const physmass massin,const double L) returns  $m_{\pi R}^{2V(6)}$  .
```

Kaon mass (theta function method):

```
double mk4Vt(const physmass massin,const double L) returns  $m_K^{2V(4)}$  .  
double mk6Vt(const physmass massin,const Li Liin,const double L) returns  $m_K^{2V(6)}$  .  
double mk6VLt(const physmass massin,const Li Liin,const double L) returns  $m_{KL}^{2V(6)}$  .  
double mk6VRt(const physmass massin,const double L) returns  $m_{KR}^{2V(6)}$  .
```

Kaon mass (Bessel function method):

```
double mk4Vb(const physmass massin,const double L) returns  $m_K^{2V(4)}$  .  
double mk6Vb(const physmass massin,const Li Liin,const double L) returns  $m_K^{2V(6)}$  .  
double mk6VLb(const physmass massin,const Li Liin,const double L) returns  $m_{KL}^{2V(6)}$  .  
double mk6VRb(const physmass massin,const double L) returns  $m_{KR}^{2V(6)}$  .
```

Eta mass (theta function method):

```
double meta4Vt(const physmass massin,const double L) returns  $m_{\eta}^{2V(4)}$  .  
double meta6Vt(const physmass massin,const Li Liin,const double L) returns  $m_{\eta}^{2V(6)}$  .  
double meta6VLt(const physmass massin,const Li Liin,const double L) returns  $m_{\eta L}^{2V(6)}$  .  
double meta6VRt(const physmass massin,const double L) returns  $m_{\eta R}^{2V(6)}$  .
```

Eta mass (Bessel function method):

```
double meta4Vb(const physmass massin,const double L) returns  $m_{\eta}^{2V(4)}$  .  
double meta6Vb(const physmass massin,const Li Liin,const double L) returns  $m_{\eta}^{2V(6)}$  .  
double meta6VLb(const physmass massin,const Li Liin,const double L) returns  $m_{\eta L}^{2V(6)}$  .  
double meta6VRb(const physmass massin,const double L) returns  $m_{\eta R}^{2V(6)}$  .
```

All these are defined in `massdecayvevV.h` and implemented in `massdecayvevV.h`. Examples of use are in `testmassdecayvevV.cc`.

8.3.2 Decay constants at finite volume

The finite volume corrections to the decay constants are defined as the difference of the decay constant in finite volume and in infinite volume:

$$\begin{aligned}\Delta^V F_a &= F_a^V - F_a^{V=\infty} = F_a^{V(4)} + F_a^{V(6)} . \\ F_a^{V(6)} &= F_{aL}^{V(6)} + F_{aR}^{V(6)} .\end{aligned}\tag{52}$$

These definitions are for $a = \pi, K, \eta$. Note that the correction is defined to the value of the decay constant, not divided by the the lowest order decay constant as in (40). The eta decay constant is defined with the octet axial current.

Pion decay constant (theta function method):

```
double fpi4Vt(const physmass massin,const double L) returns  $F_\pi^{V(4)}$ .
double fpi6Vt(const physmass massin,const Li Liin,const double L) returns  $F_\pi^{V(6)}$ .
double fpi6VLt(const physmass massin,const Li Liin,const double L) returns  $F_{\pi L}^{V(6)}$ .
double fpi6VRt(const physmass massin,const double L) returns  $F_{\pi R}^{V(6)}$ .
```

Pion decay constant (Bessel function method):

```
double fpi4Vb(const physmass massin,const double L) returns  $F_\pi^{V(4)}$ .
double fpi6Vb(const physmass massin,const Li Liin,const double L) returns  $F_\pi^{V(6)}$ .
double fpi6VLb(const physmass massin,const Li Liin,const double L) returns  $F_{\pi L}^{V(6)}$ .
double fpi6VRb(const physmass massin,const double L) returns  $F_{\pi R}^{V(6)}$ .
```

Kaon decay constant (theta function method):

```
double fk4Vt(const physmass massin,const double L) returns  $F_K^{V(4)}$ .
double fk6Vt(const physmass massin,const Li Liin,const double L) returns  $F_K^{V(6)}$ .
double fk6VLt(const physmass massin,const Li Liin,const double L) returns  $F_{KL}^{V(6)}$ .
double fk6VRt(const physmass massin,const double L) returns  $F_{KR}^{V(6)}$ .
```

Kaon decay constant (Bessel function method):

```
double fk4Vb(const physmass massin,const double L) returns  $F_K^{V(4)}$ .
double fk6Vb(const physmass massin,const Li Liin,const double L) returns  $F_K^{V(6)}$ .
double fk6VLb(const physmass massin,const Li Liin,const double L) returns  $F_{KL}^{V(6)}$ .
double fk6VRb(const physmass massin,const double L) returns  $F_{KR}^{V(6)}$ .
```

Eta decay constant (theta function method):

```
double feta4Vt(const physmass massin,const double L) returns  $F_\eta^{V(4)}$ .
double feta6Vt(const physmass massin,const Li Liin,const double L) returns  $F_\eta^{V(6)}$ .
double feta6VLt(const physmass massin,const Li Liin,const double L) returns  $F_{\eta L}^{V(6)}$ .
double feta6VRt(const physmass massin,const double L) returns  $F_{\eta R}^{V(6)}$ .
```

Eta decay constant (Bessel function method):

```
double feta4Vb(const physmass massin,const double L) returns  $F_{\eta}^{V(4)}$ .  
double feta6Vb(const physmass massin,const Li Liin,const double L) returns  $F_{\eta}^{V(6)}$ .  
double feta6VLb(const physmass massin,const Li Liin,const double L) returns  $F_{\eta L}^{V(6)}$ .  
double feta6VRb(const physmass massin,const double L) returns  $F_{\eta R}^{V(6)}$ .
```

All these are defined in `massdecayvevV.h` and implemented in `massdecayvevV.h`. Examples of use are in `testmassdecayvevV.cc`.

Acknowledgements

This work is supported, in part, by the Swedish Research Council grants 621-2011-5080 and 621-2013-4287. I thank all my collaborators in the various applications for which my version of the program made it into this collection and especially Ilaria Jemos who has tested many of the earlier versions in the course of [15].

A GNU GENERAL PUBLIC LICENSE

Version 2, June 1991

Copyright © 1989, 1991 Free Software Foundation, Inc.

51 Franklin Street, Fifth Floor, Boston, MA 02110-1301, USA

Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

Preamble

The licenses for most software are designed to take away your freedom to share and change it. By contrast, the GNU General Public License is intended to guarantee your freedom to share and change free software—to make sure the software is free for all its users. This General Public License applies to most of the Free Software Foundation’s software and to any other program whose authors commit to using it. (Some other Free Software Foundation software is covered by the GNU Library General Public License instead.) You can apply it to your programs, too.

When we speak of free software, we are referring to freedom, not price. Our General Public Licenses are designed to make sure that you have the freedom to distribute copies of free software (and charge for this service if you wish), that you receive source code or can get it if you want it, that you can change the software or use pieces of it in new free programs; and that you know you can do these things.

To protect your rights, we need to make restrictions that forbid anyone to deny you these rights or to ask you to surrender the rights. These restrictions translate to certain responsibilities for you if you distribute copies of the software, or if you modify it.

For example, if you distribute copies of such a program, whether gratis or for a fee, you must give the recipients all the rights that you have. You must make sure that they, too, receive or can get the source code. And you must show them these terms so they know their rights.

We protect your rights with two steps: (1) copyright the software, and (2) offer you this license which gives you legal permission to copy, distribute and/or modify the software.

Also, for each author's protection and ours, we want to make certain that everyone understands that there is no warranty for this free software. If the software is modified by someone else and passed on, we want its recipients to know that what they have is not the original, so that any problems introduced by others will not reflect on the original authors' reputations.

Finally, any free program is threatened constantly by software patents. We wish to avoid the danger that redistributors of a free program will individually obtain patent licenses, in effect making the program proprietary. To prevent this, we have made it clear that any patent must be licensed for everyone's free use or not licensed at all.

The precise terms and conditions for copying, distribution and modification follow.

TERMS AND CONDITIONS FOR COPYING, DISTRIBUTION AND MODIFICATION

0. This License applies to any program or other work which contains a notice placed by the copyright holder saying it may be distributed under the terms of this General Public License. The "Program", below, refers to any such program or work, and a "work based on the Program" means either the Program or any derivative work under copyright law: that is to say, a work containing the Program or a portion of it, either verbatim or with modifications and/or translated into another language. (Hereinafter, translation is included without limitation in the term "modification".) Each licensee is addressed as "you".

Activities other than copying, distribution and modification are not covered by this License; they are outside its scope. The act of running the Program is not restricted, and the output from the Program is covered only if its contents constitute a work based on the Program (independent of having been made by running the Program). Whether that is true depends on what the Program does.

1. You may copy and distribute verbatim copies of the Program's source code as you receive it, in any medium, provided that you conspicuously and appropriately publish on each copy an appropriate copyright notice and disclaimer of warranty; keep intact all the notices that refer to this License and to the absence of any warranty; and give any other recipients of the Program a copy of this License along with the Program.
You may charge a fee for the physical act of transferring a copy, and you may at your option offer warranty protection in exchange for a fee.

2. You may modify your copy or copies of the Program or any portion of it, thus forming a work based on the Program, and copy and distribute such modifications or work under the terms of Section 1 above, provided that you also meet all of these conditions:
 - (a) You must cause the modified files to carry prominent notices stating that you changed the files and the date of any change.
 - (b) You must cause any work that you distribute or publish, that in whole or in part contains or is derived from the Program or any part thereof, to be licensed as a whole at no charge to all third parties under the terms of this License.
 - (c) If the modified program normally reads commands interactively when run, you must cause it, when started running for such interactive use in the most ordinary way, to print or display an announcement including an appropriate copyright notice and a notice that there is no warranty (or else, saying that you provide a warranty) and that users may redistribute the program under these conditions, and telling the user how to view a copy of this License. (Exception: if the Program itself is interactive but does not normally print such an announcement, your work based on the Program is not required to print an announcement.)

These requirements apply to the modified work as a whole. If identifiable sections of that work are not derived from the Program, and can be reasonably considered independent and separate works in themselves, then this License, and its terms, do not apply to those sections when you distribute them as separate works. But when you distribute the same sections as part of a whole which is a work based on the Program, the distribution of the whole must be on the terms of this License, whose permissions for other licensees extend to the entire whole, and thus to each and every part regardless of who wrote it.

Thus, it is not the intent of this section to claim rights or contest your rights to work written entirely by you; rather, the intent is to exercise the right to control the distribution of derivative or collective works based on the Program.

In addition, mere aggregation of another work not based on the Program with the Program (or with a work based on the Program) on a volume of a storage or distribution medium does not bring the other work under the scope of this License.

3. You may copy and distribute the Program (or a work based on it, under Section 2) in object code or executable form under the terms of Sections 1 and 2 above provided that you also do one of the following:
 - (a) Accompany it with the complete corresponding machine-readable source code, which must be distributed under the terms of Sections 1 and 2 above on a medium customarily used for software interchange; or,

- (b) Accompany it with a written offer, valid for at least three years, to give any third party, for a charge no more than your cost of physically performing source distribution, a complete machine-readable copy of the corresponding source code, to be distributed under the terms of Sections 1 and 2 above on a medium customarily used for software interchange; or,
- (c) Accompany it with the information you received as to the offer to distribute corresponding source code. (This alternative is allowed only for noncommercial distribution and only if you received the program in object code or executable form with such an offer, in accord with Subsection b above.)

The source code for a work means the preferred form of the work for making modifications to it. For an executable work, complete source code means all the source code for all modules it contains, plus any associated interface definition files, plus the scripts used to control compilation and installation of the executable. However, as a special exception, the source code distributed need not include anything that is normally distributed (in either source or binary form) with the major components (compiler, kernel, and so on) of the operating system on which the executable runs, unless that component itself accompanies the executable.

If distribution of executable or object code is made by offering access to copy from a designated place, then offering equivalent access to copy the source code from the same place counts as distribution of the source code, even though third parties are not compelled to copy the source along with the object code.

4. You may not copy, modify, sublicense, or distribute the Program except as expressly provided under this License. Any attempt otherwise to copy, modify, sublicense or distribute the Program is void, and will automatically terminate your rights under this License. However, parties who have received copies, or rights, from you under this License will not have their licenses terminated so long as such parties remain in full compliance.
5. You are not required to accept this License, since you have not signed it. However, nothing else grants you permission to modify or distribute the Program or its derivative works. These actions are prohibited by law if you do not accept this License. Therefore, by modifying or distributing the Program (or any work based on the Program), you indicate your acceptance of this License to do so, and all its terms and conditions for copying, distributing or modifying the Program or works based on it.
6. Each time you redistribute the Program (or any work based on the Program), the recipient automatically receives a license from the original licensor to copy, distribute or modify the Program subject to these terms and conditions. You may not impose any further restrictions on the recipients' exercise of the rights granted herein. You are not responsible for enforcing compliance by third parties to this License.

7. If, as a consequence of a court judgment or allegation of patent infringement or for any other reason (not limited to patent issues), conditions are imposed on you (whether by court order, agreement or otherwise) that contradict the conditions of this License, they do not excuse you from the conditions of this License. If you cannot distribute so as to satisfy simultaneously your obligations under this License and any other pertinent obligations, then as a consequence you may not distribute the Program at all. For example, if a patent license would not permit royalty-free redistribution of the Program by all those who receive copies directly or indirectly through you, then the only way you could satisfy both it and this License would be to refrain entirely from distribution of the Program.

If any portion of this section is held invalid or unenforceable under any particular circumstance, the balance of the section is intended to apply and the section as a whole is intended to apply in other circumstances.

It is not the purpose of this section to induce you to infringe any patents or other property right claims or to contest validity of any such claims; this section has the sole purpose of protecting the integrity of the free software distribution system, which is implemented by public license practices. Many people have made generous contributions to the wide range of software distributed through that system in reliance on consistent application of that system; it is up to the author/donor to decide if he or she is willing to distribute software through any other system and a licensee cannot impose that choice.

This section is intended to make thoroughly clear what is believed to be a consequence of the rest of this License.

8. If the distribution and/or use of the Program is restricted in certain countries either by patents or by copyrighted interfaces, the original copyright holder who places the Program under this License may add an explicit geographical distribution limitation excluding those countries, so that distribution is permitted only in or among countries not thus excluded. In such case, this License incorporates the limitation as if written in the body of this License.
9. The Free Software Foundation may publish revised and/or new versions of the General Public License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns.

Each version is given a distinguishing version number. If the Program specifies a version number of this License which applies to it and “any later version”, you have the option of following the terms and conditions either of that version or of any later version published by the Free Software Foundation. If the Program does not specify a version number of this License, you may choose any version ever published by the Free Software Foundation.

10. If you wish to incorporate parts of the Program into other free programs whose distribution conditions are different, write to the author to ask for permission. For

software which is copyrighted by the Free Software Foundation, write to the Free Software Foundation; we sometimes make exceptions for this. Our decision will be guided by the two goals of preserving the free status of all derivatives of our free software and of promoting the sharing and reuse of software generally.

NO WARRANTY

11. BECAUSE THE PROGRAM IS LICENSED FREE OF CHARGE, THERE IS NO WARRANTY FOR THE PROGRAM, TO THE EXTENT PERMITTED BY APPLICABLE LAW. EXCEPT WHEN OTHERWISE STATED IN WRITING THE COPYRIGHT HOLDERS AND/OR OTHER PARTIES PROVIDE THE PROGRAM “AS IS” WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE PROGRAM IS WITH YOU. SHOULD THE PROGRAM PROVE DEFECTIVE, YOU ASSUME THE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION.
12. IN NO EVENT UNLESS REQUIRED BY APPLICABLE LAW OR AGREED TO IN WRITING WILL ANY COPYRIGHT HOLDER, OR ANY OTHER PARTY WHO MAY MODIFY AND/OR REDISTRIBUTE THE PROGRAM AS PERMITTED ABOVE, BE LIABLE TO YOU FOR DAMAGES, INCLUDING ANY GENERAL, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE USE OR INABILITY TO USE THE PROGRAM (INCLUDING BUT NOT LIMITED TO LOSS OF DATA OR DATA BEING RENDERED INACCURATE OR LOSSES SUSTAINED BY YOU OR THIRD PARTIES OR A FAILURE OF THE PROGRAM TO OPERATE WITH ANY OTHER PROGRAMS), EVEN IF SUCH HOLDER OR OTHER PARTY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

END OF TERMS AND CONDITIONS

Appendix: How to Apply These Terms to Your New Programs

If you develop a new program, and you want it to be of the greatest possible use to the public, the best way to achieve this is to make it free software which everyone can redistribute and change under these terms.

To do so, attach the following notices to the program. It is safest to attach them to the start of each source file to most effectively convey the exclusion of warranty; and each file should have at least the “copyright” line and a pointer to where the full notice is found.

one line to give the program’s name and a brief idea of what it does.
Copyright (C) yyyy name of author

This program is free software; you can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation; either version 2 of the License, or (at your option) any later version.

This program is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public License for more details.

You should have received a copy of the GNU General Public License along with this program; if not, write to the Free Software Foundation, Inc., 51 Franklin Street, Fifth Floor, Boston, MA 02110-1301, USA.

Also add information on how to contact you by electronic and paper mail.

If the program is interactive, make it output a short notice like this when it starts in an interactive mode:

```
Gnomovision version 69, Copyright (C) yyyy name of author
Gnomovision comes with ABSOLUTELY NO WARRANTY; for details type
'show w'.
```

```
This is free software, and you are welcome to redistribute it under certain
conditions; type 'show c' for details.
```

The hypothetical commands `show w` and `show c` should show the appropriate parts of the General Public License. Of course, the commands you use may be called something other than `show w` and `show c`; they could even be mouse-clicks or menu items—whatever suits your program.

You should also get your employer (if you work as a programmer) or your school, if any, to sign a “copyright disclaimer” for the program, if necessary. Here is a sample; alter the names:

```
Yoyodyne, Inc., hereby disclaims all copyright interest in the program
'Gnomovision' (which makes passes at compilers) written by James Hacker.
```

```
signature of Ty Coon, 1 April 1989
Ty Coon, President of Vice
```

This General Public License does not permit incorporating your program into proprietary programs. If your program is a subroutine library, you may consider it more useful to permit linking proprietary applications with the library. If this is what you want to do, use the GNU Library General Public License instead of this License.

B Creative Commons Attribution 4.0 International Public License

By exercising the Licensed Rights (defined below), You accept and agree to be bound by the terms and conditions of this Creative Commons Attribution 4.0 International Public License ("Public License"). To the extent this Public License may be interpreted as a contract, You are granted the Licensed Rights in consideration of Your acceptance of these terms and conditions, and the Licensor grants You such rights in consideration of benefits the Licensor receives from making the Licensed Material available under these terms and conditions.

Section 1 – Definitions.

- a Adapted Material** means material subject to Copyright and Similar Rights that is derived from or based upon the Licensed Material and in which the Licensed Material is translated, altered, arranged, transformed, or otherwise modified in a manner requiring permission under the Copyright and Similar Rights held by the Licensor. For purposes of this Public License, where the Licensed Material is a musical work, performance, or sound recording, Adapted Material is always produced where the Licensed Material is synched in timed relation with a moving image.
- b Adapter's License** means the license You apply to Your Copyright and Similar Rights in Your contributions to Adapted Material in accordance with the terms and conditions of this Public License.
- c Copyright and Similar Rights** means copyright and/or similar rights closely related to copyright including, without limitation, performance, broadcast, sound recording, and Sui Generis Database Rights, without regard to how the rights are labeled or categorized. For purposes of this Public License, the rights specified in Section 2(b)(1)-(2) are not Copyright and Similar Rights.
- d Effective Technological Measures** means those measures that, in the absence of proper authority, may not be circumvented under laws fulfilling obligations under Article 11 of the WIPO Copyright Treaty adopted on December 20, 1996, and/or similar international agreements.
- e Exceptions and Limitations** means fair use, fair dealing, and/or any other exception or limitation to Copyright and Similar Rights that applies to Your use of the Licensed Material.
- f Licensed Material** means the artistic or literary work, database, or other material to which the Licensor applied this Public License.
- g Licensed Rights** means the rights granted to You subject to the terms and conditions of this Public License, which are limited to all Copyright and Similar Rights that apply to Your use of the Licensed Material and that the Licensor has authority to license.
- h Licensor** means the individual(s) or entity(ies) granting rights under this Public License.
- i Share** means to provide material to the public by any means or process that requires permission under the Licensed Rights, such as reproduction, public display, public performance, distribution, dissemination, communication, or importation, and to make material available

to the public including in ways that members of the public may access the material from a place and at a time individually chosen by them.

j Sui Generis Database Rights means rights other than copyright resulting from Directive 96/9/EC of the European Parliament and of the Council of 11 March 1996 on the legal protection of databases, as amended and/or succeeded, as well as other essentially equivalent rights anywhere in the world.

j You means the individual or entity exercising the Licensed Rights under this Public License. **Your** has a corresponding meaning.

Section 2 – Scope.

a License grant.

1. Subject to the terms and conditions of this Public License, the Licensor hereby grants You a worldwide, royalty-free, non-sublicensable, non-exclusive, irrevocable license to exercise the Licensed Rights in the Licensed Material to:
 - A. reproduce and Share the Licensed Material, in whole or in part; and
 - B. produce, reproduce, and Share Adapted Material.
2. Exceptions and Limitations. For the avoidance of doubt, where Exceptions and Limitations apply to Your use, this Public License does not apply, and You do not need to comply with its terms and conditions.
3. Term. The term of this Public License is specified in Section 6(a).
4. Media and formats; technical modifications allowed. The Licensor authorizes You to exercise the Licensed Rights in all media and formats whether now known or hereafter created, and to make technical modifications necessary to do so. The Licensor waives and/or agrees not to assert any right or authority to forbid You from making technical modifications necessary to exercise the Licensed Rights, including technical modifications necessary to circumvent Effective Technological Measures. For purposes of this Public License, simply making modifications authorized by this Section 2(a)(4) never produces Adapted Material.
5. Downstream recipients.
 - A. Offer from the Licensor – Licensed Material. Every recipient of the Licensed Material automatically receives an offer from the Licensor to exercise the Licensed Rights under the terms and conditions of this Public License.
 - B. No downstream restrictions. You may not offer or impose any additional or different terms or conditions on, or apply any Effective Technological Measures to, the Licensed Material if doing so restricts exercise of the Licensed Rights by any recipient of the Licensed Material.
6. No endorsement. Nothing in this Public License constitutes or may be construed as permission to assert or imply that You are, or that Your use of the Licensed Material is, connected with, or sponsored, endorsed, or granted official status by, the Licensor or others designated to receive attribution as provided in Section 3(a)(1)(A)(i).

b Other rights.

1. Moral rights, such as the right of integrity, are not licensed under this Public License, nor are publicity, privacy, and/or other similar personality rights; however, to the extent possible, the Licensor waives and/or agrees not to assert any such rights held by the Licensor to the limited extent necessary to allow You to exercise the Licensed Rights, but not otherwise.
2. Patent and trademark rights are not licensed under this Public License.
3. To the extent possible, the Licensor waives any right to collect royalties from You for the exercise of the Licensed Rights, whether directly or through a collecting society under any voluntary or waivable statutory or compulsory licensing scheme. In all other cases the Licensor expressly reserves any right to collect such royalties.

Section 3 – License Conditions.

Your exercise of the Licensed Rights is expressly made subject to the following conditions.

a. Attribution.

1. If You Share the Licensed Material (including in modified form), You must:
 - A.** retain the following if it is supplied by the Licensor with the Licensed Material:
 - i. identification of the creator(s) of the Licensed Material and any others designated to receive attribution, in any reasonable manner requested by the Licensor (including by pseudonym if designated);
 - ii. a copyright notice;
 - iii. a notice that refers to this Public License;
 - iv. a notice that refers to the disclaimer of warranties;
 - v. a URI or hyperlink to the Licensed Material to the extent reasonably practicable;
 - B.** indicate if You modified the Licensed Material and retain an indication of any previous modifications; and
 - C.** indicate the Licensed Material is licensed under this Public License, and include the text of, or the URI or hyperlink to, this Public License.
2. You may satisfy the conditions in Section 3(a)(1) in any reasonable manner based on the medium, means, and context in which You Share the Licensed Material. For example, it may be reasonable to satisfy the conditions by providing a URI or hyperlink to a resource that includes the required information.
3. If requested by the Licensor, You must remove any of the information required by Section 3(a)(1)(A) to the extent reasonably practicable.
4. If You Share Adapted Material You produce, the Adapter's License You apply must not prevent recipients of the Adapted Material from complying with this Public License.

Section 4 – Sui Generis Database Rights.

Where the Licensed Rights include Sui Generis Database Rights that apply to Your use of the Licensed Material:

- a. for the avoidance of doubt, Section 2(a)(1) grants You the right to extract, reuse, reproduce, and Share all or a substantial portion of the contents of the database;
- b. if You include all or a substantial portion of the database contents in a database in which You have Sui Generis Database Rights, then the database in which You have Sui Generis Database Rights (but not its individual contents) is Adapted Material; and
- c. You must comply with the conditions in Section 3(a) if You Share all or a substantial portion of the contents of the database.

For the avoidance of doubt, this Section 4 supplements and does not replace Your obligations under this Public License where the Licensed Rights include other Copyright and Similar Rights.

Section 5 – Disclaimer of Warranties and Limitation of Liability.

- a. Unless otherwise separately undertaken by the Licensor, to the extent possible, the Licensor offers the Licensed Material as-is and as-available, and makes no representations or warranties of any kind concerning the Licensed Material, whether express, implied, statutory, or other. This includes, without limitation, warranties of title, merchantability, fitness for a particular purpose, non-infringement, absence of latent or other defects, accuracy, or the presence or absence of errors, whether or not known or discoverable. Where disclaimers of warranties are not allowed in full or in part, this disclaimer may not apply to You.
- b. To the extent possible, in no event will the Licensor be liable to You on any legal theory (including, without limitation, negligence) or otherwise for any direct, special, indirect, incidental, consequential, punitive, exemplary, or other losses, costs, expenses, or damages arising out of this Public License or use of the Licensed Material, even if the Licensor has been advised of the possibility of such losses, costs, expenses, or damages. Where a limitation of liability is not allowed in full or in part, this limitation may not apply to You.
- c. The disclaimer of warranties and limitation of liability provided above shall be interpreted in a manner that, to the extent possible, most closely approximates an absolute disclaimer and waiver of all liability.

Section 6 – Term and Termination.

- a. This Public License applies for the term of the Copyright and Similar Rights licensed here. However, if You fail to comply with this Public License, then Your rights under this Public License terminate automatically.
- b. Where Your right to use the Licensed Material has terminated under Section 6(a), it reinstates:

1. automatically as of the date the violation is cured, provided it is cured within 30 days of Your discovery of the violation; or
2. upon express reinstatement by the Licensor.

For the avoidance of doubt, this Section 6(b) does not affect any right the Licensor may have to seek remedies for Your violations of this Public License.

- c. For the avoidance of doubt, the Licensor may also offer the Licensed Material under separate terms or conditions or stop distributing the Licensed Material at any time; however, doing so will not terminate this Public License.
- d. Sections 1, 5, 6, 7, and 8 survive termination of this Public License.

Section 7 – Other Terms and Conditions.

- a. The Licensor shall not be bound by any additional or different terms or conditions communicated by You unless expressly agreed.
- b. Any arrangements, understandings, or agreements regarding the Licensed Material not stated herein are separate from and independent of the terms and conditions of this Public License.

Section 8 – Interpretation.

- a. For the avoidance of doubt, this Public License does not, and shall not be interpreted to, reduce, limit, restrict, or impose conditions on any use of the Licensed Material that could lawfully be made without permission under this Public License.
- b. To the extent possible, if any provision of this Public License is deemed unenforceable, it shall be automatically reformed to the minimum extent necessary to make it enforceable. If the provision cannot be reformed, it shall be severed from this Public License without affecting the enforceability of the remaining terms and conditions.
- c. No term or condition of this Public License will be waived and no failure to comply consented to unless expressly agreed to by the Licensor.
- d. Nothing in this Public License constitutes or may be interpreted as a limitation upon, or waiver of, any privileges and immunities that apply to the Licensor or You, including from the legal processes of any jurisdiction or authority.

References

- [1] J. Bijnens, arXiv:1412.0887 [hep-ph]. XXX add published XXX
- [2] <http://creativecommons.org/licenses/by/4.0/>.
- [3] <http://en.wikipedia.org/wiki/Chiron>
- [4] <http://www.gnu.org/licenses/gpl-2.0.html><http://www.gnu.org/licenses/gpl-2.0.html>
- [5] <http://www.thep.lu.se/~bijnens/chiron/>

- [6] G. 't Hooft, M. J. G. Veltman, Nucl. Phys. **B153** (1979) 365-401.
- [7] <http://cernlib.web.cern.ch>
- [8] J. Bijnens, E. Boström and T. A. Lähde, JHEP **1401** (2014) 019 [arXiv:1311.3531 [hep-lat]].
- [9] A. van Doren and L. de Ridder, J. Comput. Appl. Math. **2** (1976) 207-217.
- [10] S. Weinberg, Physica A **96** (1979) 327.
- [11] J. Gasser and H. Leutwyler, Annals Phys. **158** (1984) 142.
- [12] J. Gasser and H. Leutwyler, Nucl. Phys. B **250** (1985) 465.
- [13] <http://www.thep.lu.se/~bijnens/chpt/>
- [14] J. Bijnens, Prog. Part. Nucl. Phys. **58** (2007) 521 [hep-ph/0604043].
- [15] J. Bijnens and I. Jemos, Nucl. Phys. B **854** (2012) 631 [arXiv:1103.5945 [hep-ph]].
- [16] J. Bijnens and G. Ecker, arXiv:1405.6488 [hep-ph]. XXX add published XXX
- [17] J. Bijnens, G. Colangelo and G. Ecker, JHEP **9902** (1999) 020 [hep-ph/9902437].
- [18] J. Bijnens, G. Colangelo and G. Ecker, Annals Phys. **280** (2000) 100 [hep-ph/9907333].
- [19] G. Amorós, J. Bijnens and P. Talavera, Nucl. Phys. B **568** (2000) 319 [hep-ph/9907264].
- [20] J. Bijnens and P. Talavera, JHEP **0203** (2002) 046 [hep-ph/0203049].
- [21] G. Passarino and M. J. G. Veltman, Nucl. Phys. B **160** (1979) 151.
- [22] J. Bijnens, N. Danielsson and T. A. Lähde, Phys. Rev. D **73** (2006) 074509 [hep-lat/0602003].
- [23] J. Bijnens and T. A. Lähde, Phys. Rev. D **72** (2005) 074502 [hep-lat/0506004].
- [24] J. Bijnens and T. A. Lähde, Phys. Rev. D **71** (2005) 094502 [hep-lat/0501014].
- [25] J. Bijnens, N. Danielsson and T. A. Lähde, Phys. Rev. D **70** (2004) 111503 [hep-lat/0406017].
- [26] J. Bijnens and T. Rössler, arXiv:1411.6384 [hep-lat]. XXX add published XXX
- [27] J. Bijnens and K. Ghorbani, Phys. Lett. B **636** (2006) 51 [hep-lat/0602019].
- [28] E. Golowich and J. Kambor, Phys. Rev. D **58** (1998) 036004 [hep-ph/9710214].
- [29] G. Amorós, J. Bijnens and P. Talavera, Nucl. Phys. B **585** (2000) 293 [Erratum-ibid. B **598** (2001) 665] [hep-ph/0003258].

Index

+, 19, 20
−, 19, 20
*, 19, 20
<<, 16–20
==, 16–18
>>, 16–20
A22bVb, 28
A22bVt, 28
A23bVb, 28
A23bVt, 28
AbVb, 28
AbVt, 28
Abeps, 22
Ab, 21, 22
B1bnum, 23
B1b, 23
B21bnum, 23
B21b, 23
B22bnum, 23
B22b, 23
B31bnum, 23
B32bnum, 23
BbVb, 28
BbVt, 28
Bbeps, 22
Bbnum, 23
Bb, 21, 23
Cbeps, 22
Cb, 21
CirandomlargeNc2, 20
CirandomlargeNc, 20
Cirandom, 20
Ci, 19
DINTEGRAL, 25, 27, 28
LiCiBE14.dat, 19
LirandomlargeNc2, 19
LirandomlargeNc, 19
Lirandom, 19
Li, 18
SINTEGRAL, 25
WINTEGRAL, 23
changescale, 18, 20
dcomplex, 6
feta4Llo, 40
feta4L, 36
feta4Rlo, 40
feta4R, 36
feta4Vb, 45
feta4Vt, 44
feta4lo, 40
feta4, 36
feta6Clo, 41
feta6C, 36
feta6Llo, 41
feta6L, 36
feta6Rlo, 41
feta6R, 36
feta6VLb, 45
feta6VLt, 44
feta6VRb, 45
feta6VRt, 44
feta6Vb, 45
feta6Vt, 44
feta6lo, 41
feta6, 36
finitevolumeoneloopintegrals.cc, 29
finitevolumeoneloopintegrals.h, 29
finitevolumesunsetintegrals.cc, 31, 34
finitevolumesunsetintegrals.h, 31, 34
fk4Llo, 40
fk4L, 36
fk4Rlo, 40
fk4R, 36
fk4Vb, 44
fk4Vt, 44
fk4lo, 40
fk4, 36
fk6Clo, 40
fk6C, 36
fk6Llo, 40
fk6L, 36
fk6Rlo, 40

fk6R, 36
 fk6VLb, 44
 fk6VLt, 44
 fk6VRb, 44
 fk6VRt, 44
 fk6Vb, 44
 fk6Vt, 44
 fk6lo, 40
 fk6, 36
 fpi4Llo, 40
 fpi4L, 36
 fpi4Rlo, 40
 fpi4R, 36
 fpi4Vb, 44
 fpi4Vt, 44
 fpi4lo, 40
 fpi4, 36
 fpi6Clo, 40
 fpi6C, 36
 fpi6Llo, 40
 fpi6L, 36
 fpi6Rlo, 40
 fpi6R, 36
 fpi6VLb, 44
 fpi6VLt, 44
 fpi6VRb, 44
 fpi6VRt, 44
 fpi6Vb, 44
 fpi6Vt, 44
 fpi6lo, 40
 fpi6, 36
 getB0mhat, 18
 getB0ms, 18
 getf0, 17, 18
 getfpimeta.cc, 37
 getfpimeta.h, 37
 getfpimeta4, 37
 getfpimeta6, 37
 getfpimeta, 36
 getfpi, 16
 getmeta, 16
 getmk0, 17
 getmk, 16
 getmp0, 17
 getmpi, 16
 getmu, 16–18
 getprecisiononeloopintegrals, 24
 getprecisionquenchedsunsetintegrals, 27
 getprecisionsunsetintegrals, 26
 hh1Vb, 31, 33
 hh1Vt, 30, 32
 hh1dVb, 31, 33
 hh1dVt, 30, 33
 hh1d, 25, 27
 hh1, 25, 27
 hh21Vb, 31, 33
 hh21Vt, 30, 32
 hh21dVb, 31, 33
 hh21dVt, 30, 33
 hh21d, 25, 27
 hh21, 25, 27
 hh22Vb, 31, 33
 hh22Vt, 30, 32
 hh22dVb, 31, 33
 hh22dVt, 30, 33
 hh27Vb, 31, 33
 hh27Vt, 30, 33
 hh27dVb, 31, 33
 hh27dVt, 30, 33
 hh31, 25
 hhVb, 31, 33
 hhVt, 30, 32
 hhdVb, 31, 33
 hhdVt, 30, 33
 hhd, 25, 27
 hh, 25, 27
 jbdad2, 14
 jbdad3, 15
 jbdbesi0, 6
 jbdbesi1, 6
 jbdbesk0, 7
 jbdbesk1, 7
 jbdbesk2, 7
 jbdbesk3, 7
 jbdbesk4, 7
 jbdcauch2, 12
 jbdcauch, 11
 jbdgauss2, 10
 jbdgauss, 10
 jbdli2, 6

jbdquad15, 10	mk4L, 35
jbdquad21, 11	mk4Rlo, 39
jbdsing15, 12	mk4R, 35
jbdsing21, 12	mk4Vb, 43
jbdtheta2d02, 9	mk4Vt, 43
jbdtheta2d0m1, 9	mk4lo, 39
jbdtheta2d0, 9	mk4, 35
jbdtheta30m1, 8	mk6Clo, 39
jbdtheta30, 7	mk6C, 35
jbdtheta32, 8	mk6Llo, 39
jbdtheta34, 8	mk6L, 35
jbnunlib, 6	mk6Rlo, 39
jbwgauss2, 13	mk6R, 35
jbwgauss, 13	mk6VLb, 43
jbwquad15, 14	mk6VLt, 43
jbwquad21, 14	mk6VRb, 43
lomass, 16	mk6VRt, 43
massdecayvevV.h, 43, 45	mk6Vb, 43
massesdecayvev.cc, 35, 36, 38	mk6Vt, 43
massesdecayvev.h, 35, 36, 38	mk6lo, 39
massesdecayvevlo.cc, 39, 41, 42	mk6, 35
massesdecayvevlo.h, 39, 41, 42	mpi4Llo, 39
meta4Llo, 39	mpi4L, 34
meta4L, 35	mpi4Rlo, 39
meta4Rlo, 39	mpi4R, 34
meta4R, 35	mpi4Vb, 43
meta4Vb, 43	mpi4Vt, 43
meta4Vt, 43	mpi4lo, 39
meta4lo, 39	mpi4, 34
meta4, 35	mpi6Clo, 39
meta6Clo, 39	mpi6C, 34
meta6C, 35	mpi6Llo, 39
meta6Llo, 39	mpi6L, 34
meta6L, 35	mpi6Rlo, 39
meta6Rlo, 39	mpi6R, 35
meta6R, 35	mpi6VLb, 43
meta6VLb, 43	mpi6VLt, 43
meta6VLt, 43	mpi6VRb, 43
meta6VRb, 43	mpi6VRt, 43
meta6VRt, 43	mpi6Vb, 43
meta6Vb, 43	mpi6Vt, 43
meta6Vt, 43	mpi6lo, 39
meta6lo, 39	mpi6, 34
meta6, 35	oneloopintegrals.cc, 22–24
mk4Llo, 39	oneloopintegrals.h, 22–24

- out, 16–18, 20
- physmass, 16
- qqstrange4Llo, 42
- qqstrange4L, 38
- qqstrange4Rlo, 42
- qqstrange4R, 38
- qqstrange4lo, 42
- qqstrange4, 38
- qqstrange6Clo, 42
- qqstrange6C, 38
- qqstrange6Llo, 42
- qqstrange6L, 38
- qqstrange6Rlo, 42
- qqstrange6R, 38
- qqstrange6lo, 42
- qqstrange6, 38
- qqup4Llo, 41
- qqup4L, 38
- qqup4Rlo, 41
- qqup4R, 38
- qqup4lo, 41
- qqup4, 38
- qqup6Clo, 42
- qqup6C, 38
- qqup6Llo, 41
- qqup6L, 38
- qqup6Rlo, 42
- qqup6R, 38
- qqup6lo, 41
- qqup6, 38
- quarkmass, 17
- quenchedsunsetintegrals.cc, 27
- quenchedsunsetintegrals.h, 27
- setB0mhat, 17
- setB0ms, 17
- setci, 19
- setf0, 17
- setfpi, 16
- setli, 18
- setmeta, 16
- setmk0, 17
- setmk, 16
- setmp0, 17
- setmpi, 16
- setmu, 16–19

- setname, 18, 19
- setprecisionfinitevolumeoneloopb, 29
- setprecisionfinitevolumeoneloopt, 29
- setprecisionfinitevolumesunsetb, 31, 33
- setprecisionfinitevolumesunsett, 31, 33
- setprecisiononeloopintegrals, 24
- setprecisionquenchedsunsetintegrals, 27
- setprecisionsunsetintegrals, 25
- sunsetintegrals.cc, 26
- sunsetintegrals.h, 26
- testfinitevolumeoneloopintegrals.cc, 29
- testfinitevolumesunsetintegrals.cc, 31, 34
- testgetfpimeta.cc, 37
- testintegralscomplex.cc, 13
- testintegralsreal.cc, 10
- testintegralsrealsingular.cc, 11
- testmassdecayvev.cc, 35, 36
- testmassdecayvevV.cc, 43, 45
- testmassdecayvevlo.cc, 39, 41
- testmassesdecayvev.cc, 38
- testmassesdecayvevlo.cc, 42
- testoneloopintegrals.cc, 22–24
- testquenchedsunsetintegrals.cc, 27
- testsunsetintegrals.cc, 26
- zhh1d, 25
- zhh1, 25
- zhh21d, 25
- zhh21, 25
- zhh31, 25
- zhhd, 25
- zhh, 25
- Bubbles, 22
- Files, 5
- Guidelines, 4
- Installation, 5
- Introduction, 3
- Loop integrals, 20
- Quenched sunsets, 26
- Sunsets, 24
- Tadpoles, 21
- Testroutines, 6